

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	2
1.1	DataFrame import and filters	2
1.2	Metadata	4
2	Model(s) analysis	5
2.1	Explanation legend tables	5
2.2	Repartition of train and test	6
2.2.1	External Split: X_train vs X_test	6
2.2.2	Internal CV Folds	6
2.3	Model : Linear	8
2.3.1	Gridsearch	8
2.3.2	Scatter	10
2.3.3	Scatter residuals	11
2.3.4	Learning curve	11
2.4	Model : XGB	12
2.4.1	Gridsearch	12
2.4.2	Scatter	18
2.4.3	Scatter residuals	19
2.4.4	Learning curve	19
2.5	Model : RFR	20
2.5.1	Gridsearch	20
2.5.2	Scatter	25
2.5.3	Scatter residuals	26
2.5.4	Learning curve	26
2.6	Sumup-table	27
3	Plot	27
3.1	Standalone plots	27
3.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	27
3.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	30

3.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	33
3.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	36
3.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	39
3.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	42
3.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	45
3.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	48
3.2	Compare plots	51
3.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	51
3.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	53
3.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	54
3.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	55
3.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	57
3.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	58
3.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	59
3.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	61

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
```

```
#df['campaign'] = "ALL"
```

```
return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_NH3_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl4", 6: "sl3", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl4", 3: "sl3", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

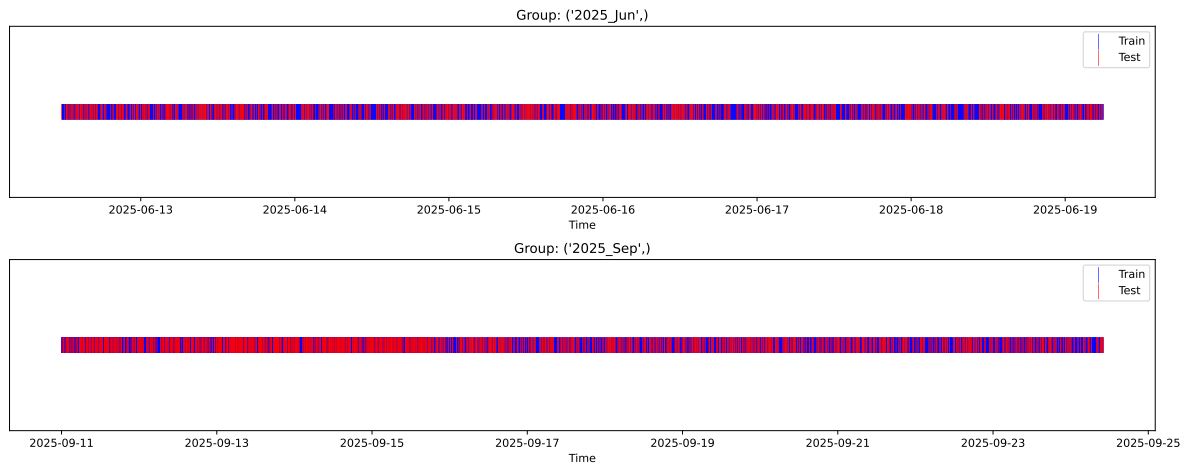
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

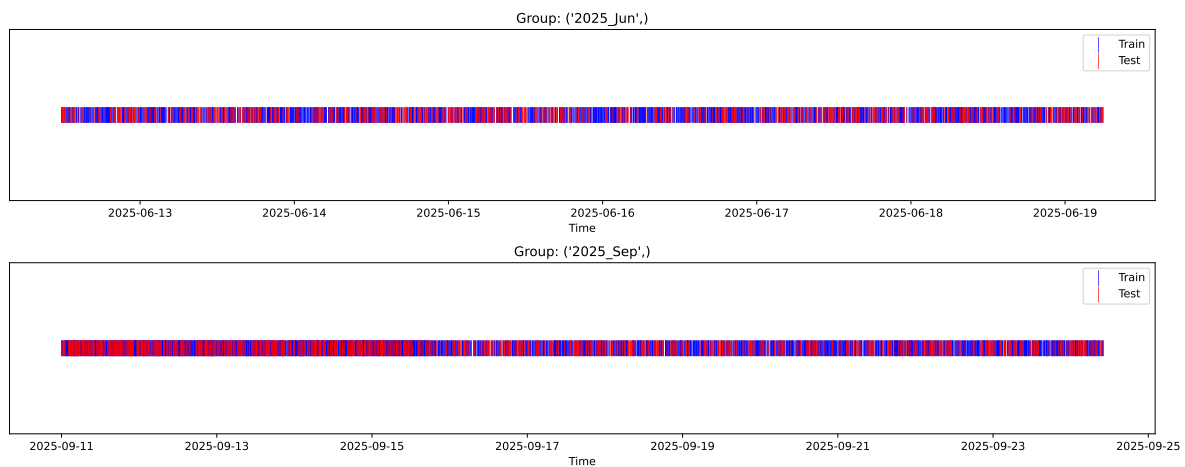
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

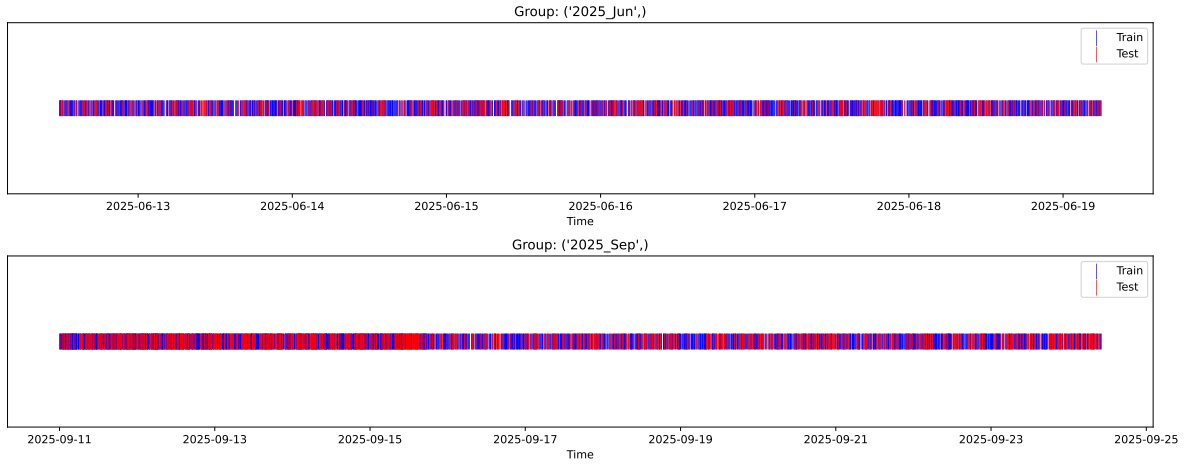
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



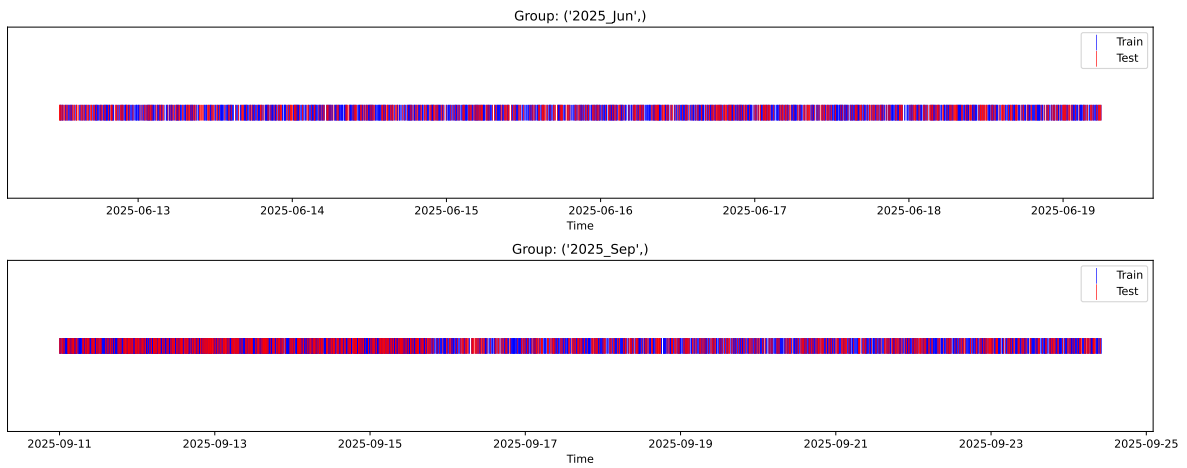
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



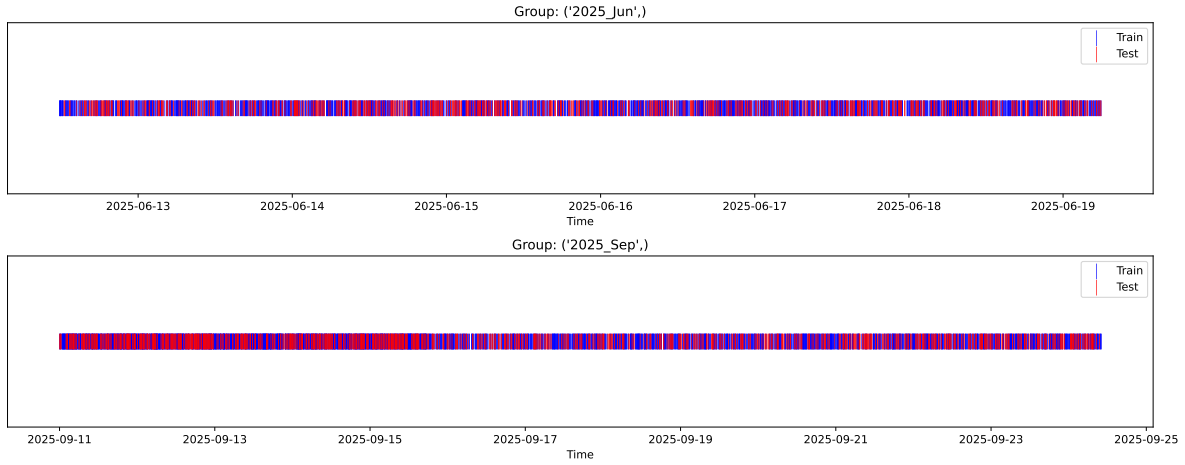
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



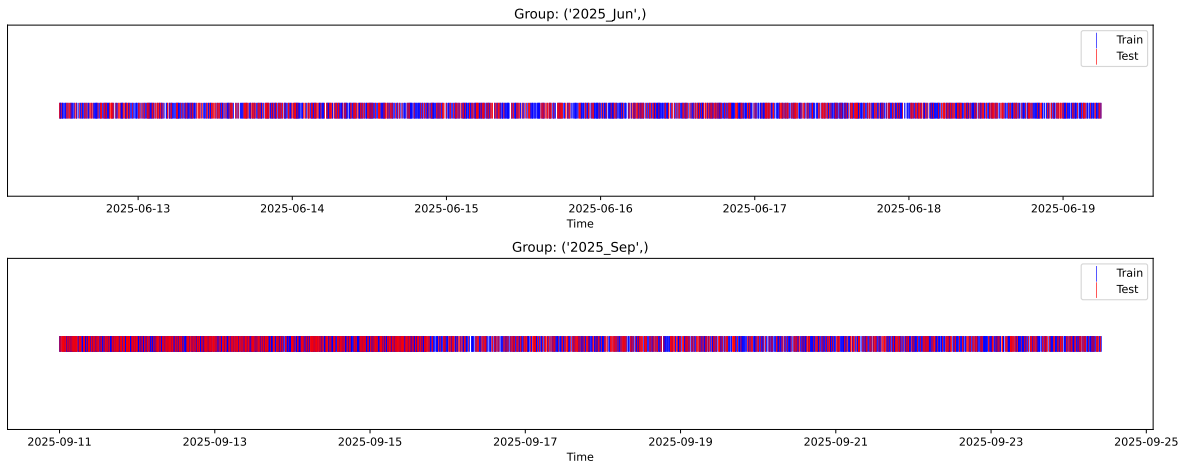
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.339, std=0.798, min=0.959, max=13.845
 y_{test} : mean=2.374, std=0.786, min=1.071, max=7.277

Fold 0 — y_{test} : mean=2.34, std=0.82, y_{train} : mean=2.34, std=0.79

Fold 1 — y_{test} : mean=2.34, std=0.77, y_{train} : mean=2.34, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.85, y_{train} : mean=2.33, std=0.78

Fold 3 — y_test: mean=2.36, std=0.84, y_train: mean=2.33, std=0.78

Fold 4 — y_test: mean=2.35, std=0.76, y_train: mean=2.34, std=0.81

train scores CV: [0.35126041 0.34129782 0.38941971 0.36572649 0.35147845]

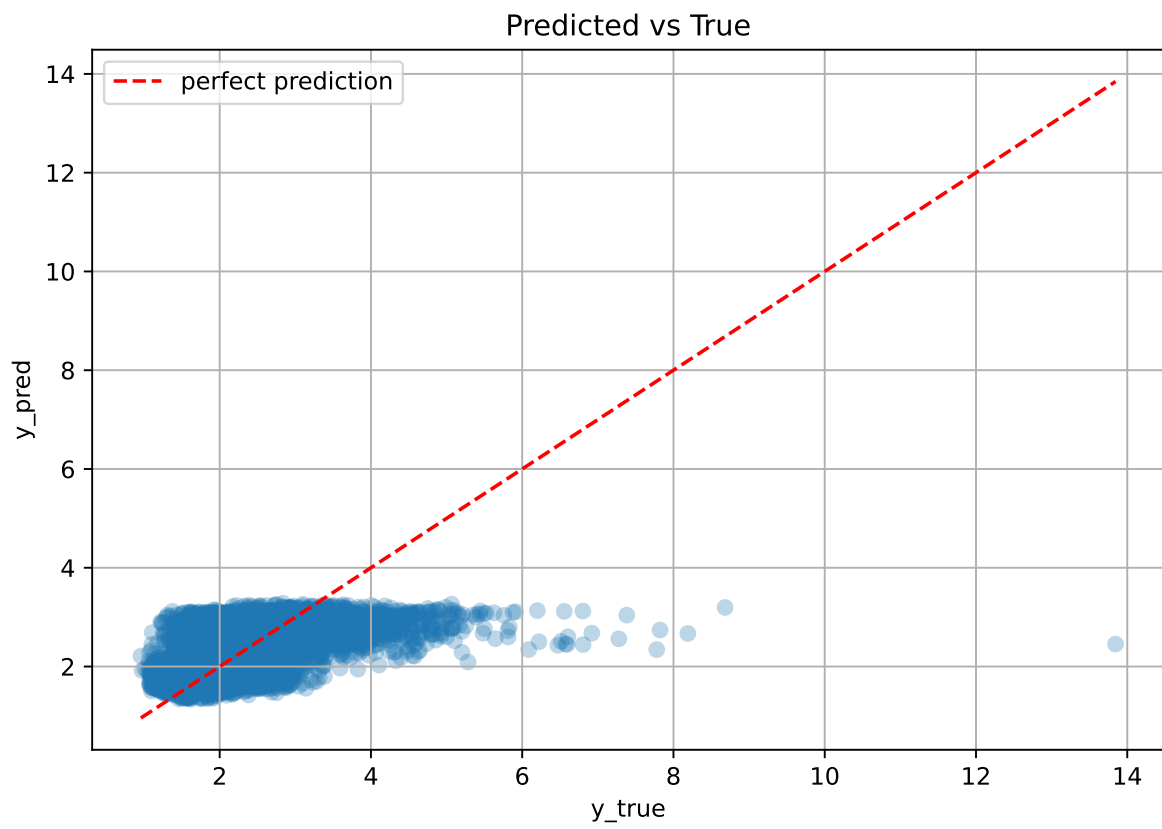
test scores CV: [0.37881413 0.40998904 0.30281638 0.34887505 0.38406497]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3598	0.3649	0.0366	0.3451	0.3453	-0.0197	0.9461	0.9861

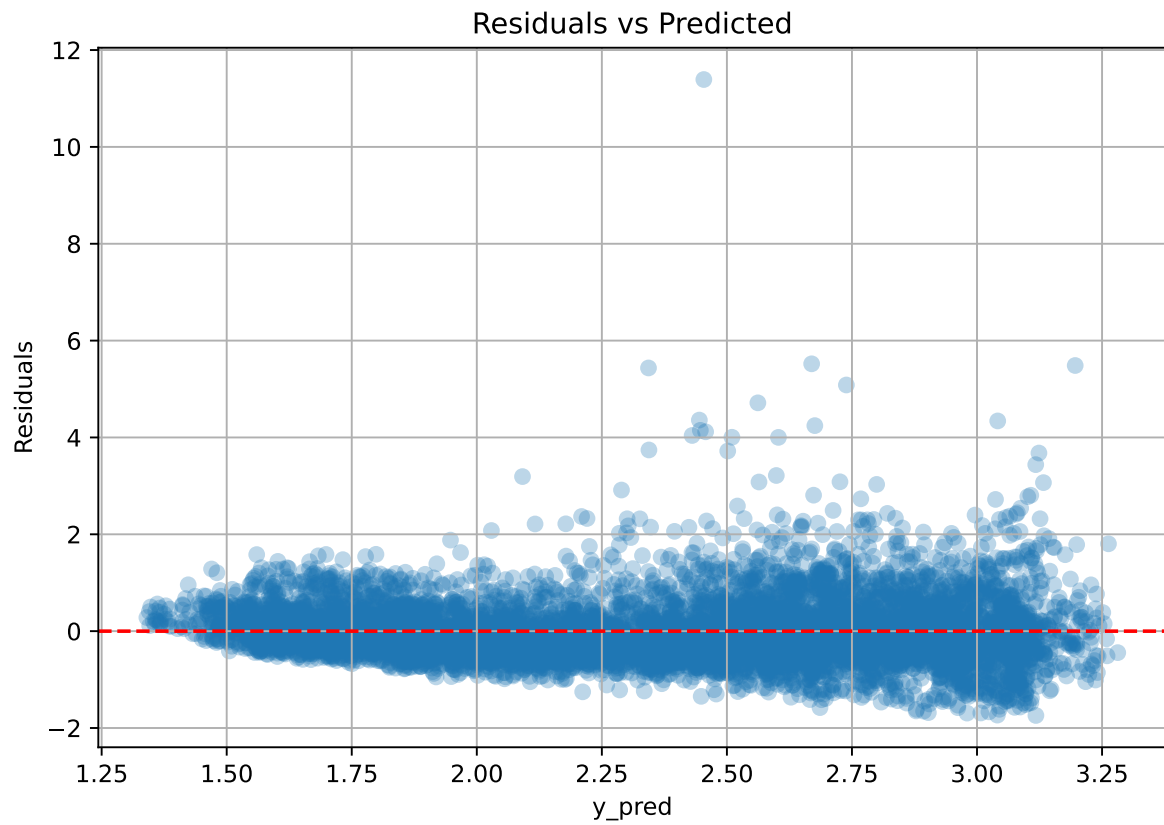
Coefficients:

coef_r_CO2 = -0.0491, coef_r_NH3 = -0.0447, coef_r_THI = 0.5970, coef_r_temperature = -0.7372, coef_r_umidity = -0.6671, intercept = 2.3392,

2.3.2 Scatter



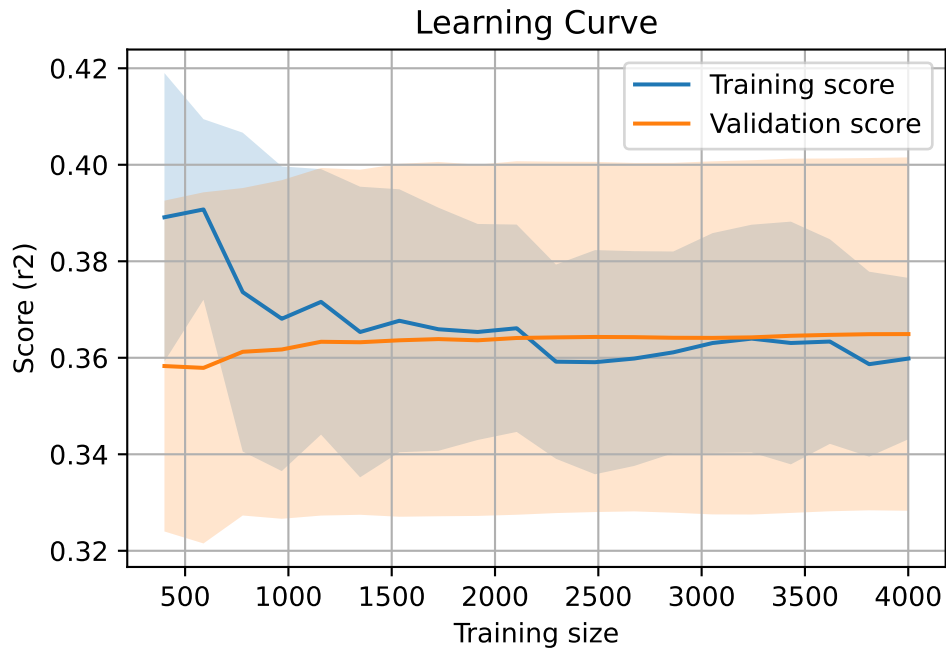
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

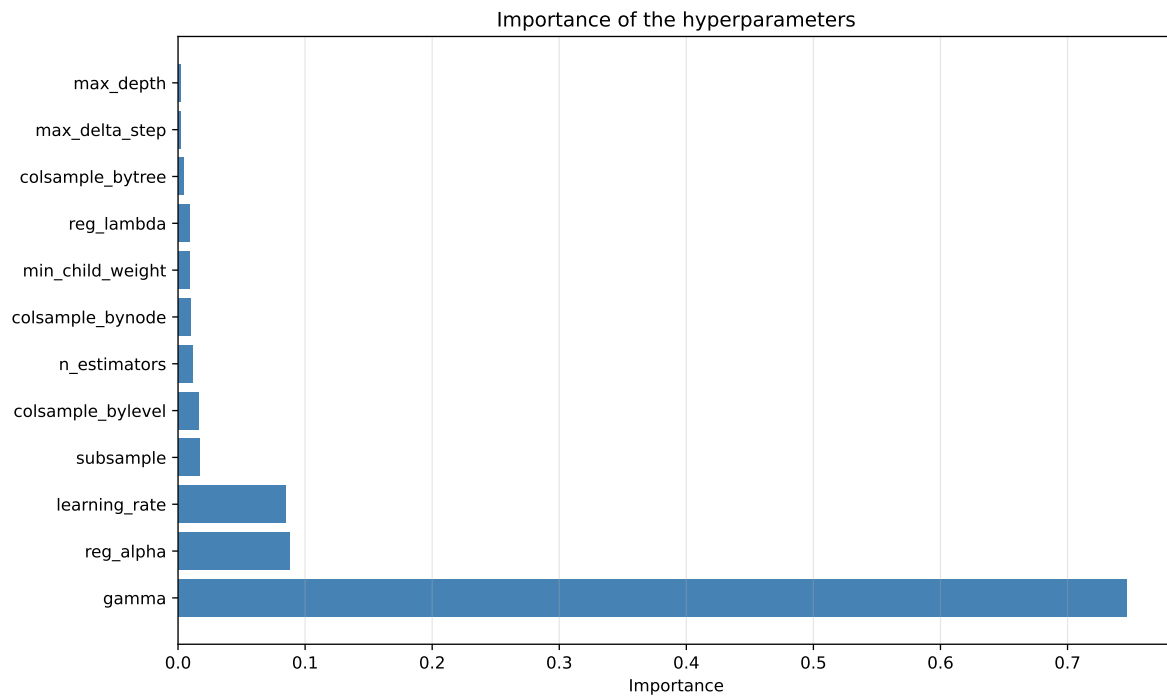
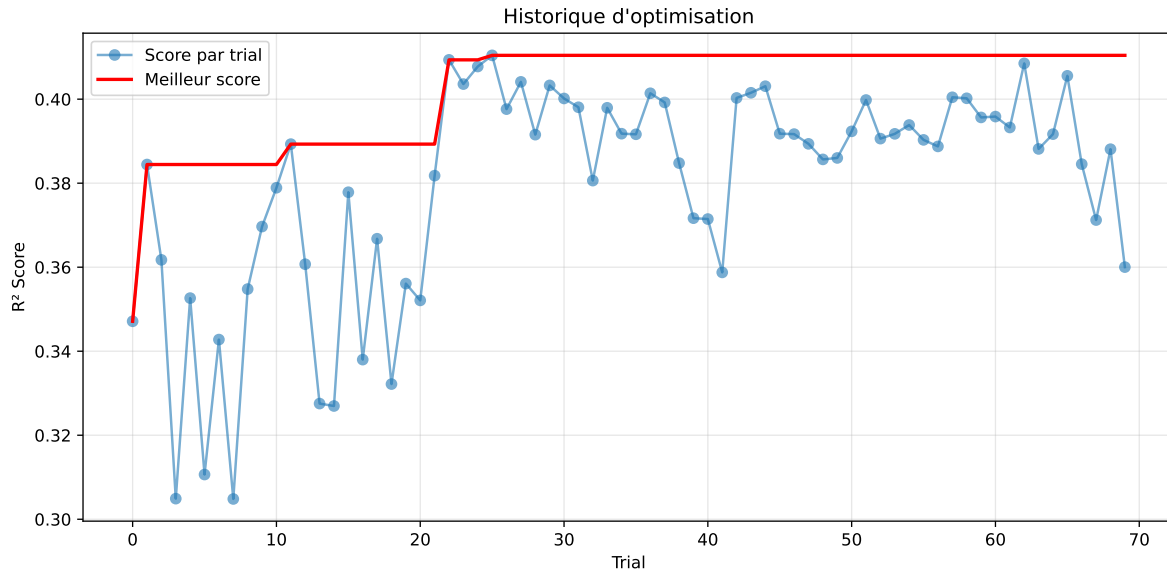


2.4 Model : XGB

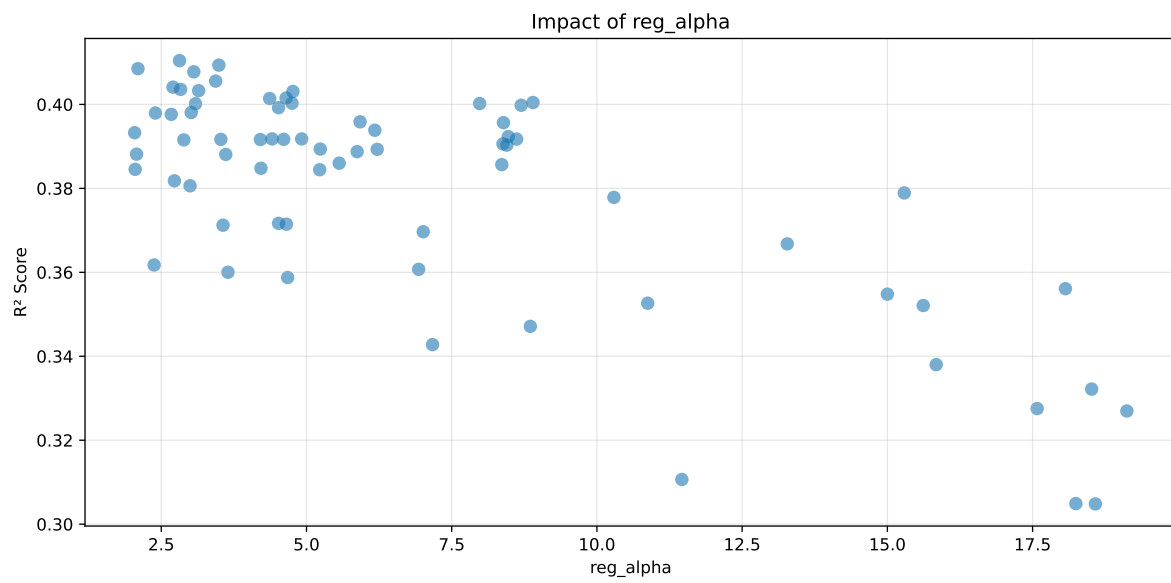
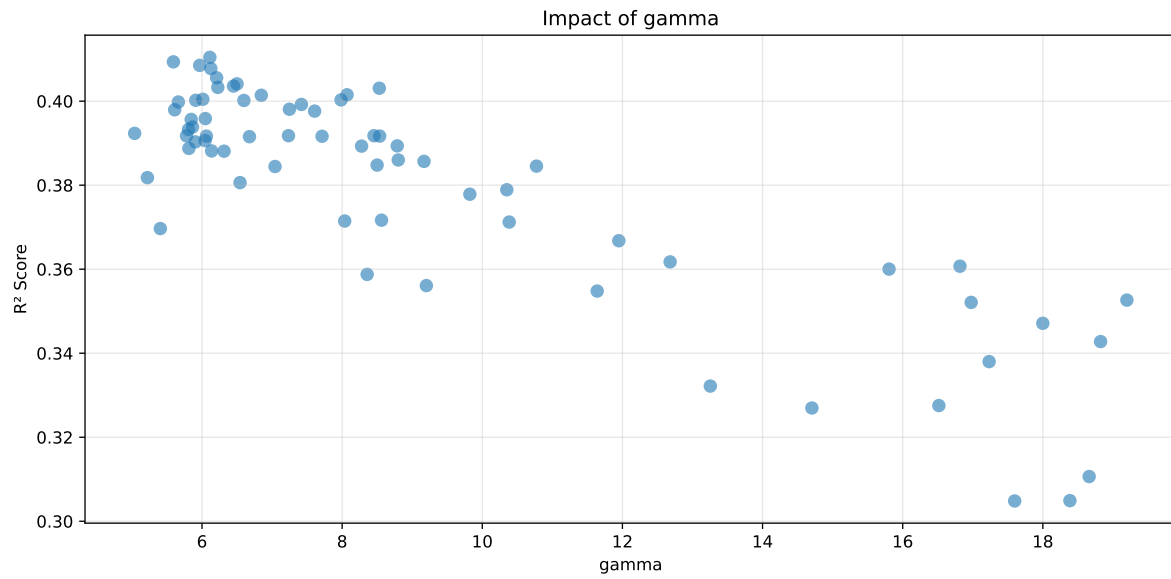
2.4.1 Gridsearch

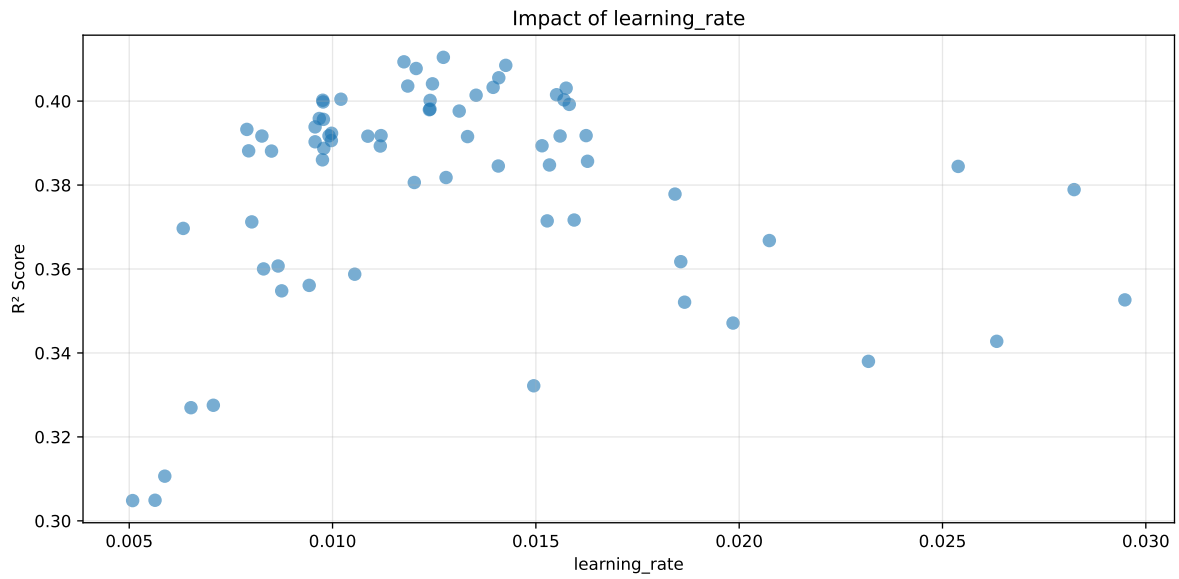
Distribution y_{train} vs y_{test} : y_{train} : mean=2.339, std=0.798, min=0.959, max=13.845
 y_{test} : mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====
 feature importance $r_{umidity}$ 0.322744 $r_{temperature}$ 0.251915 r_{THI} 0.215098 r_{NH3} 0.141881 r_{CO2} 0.068362



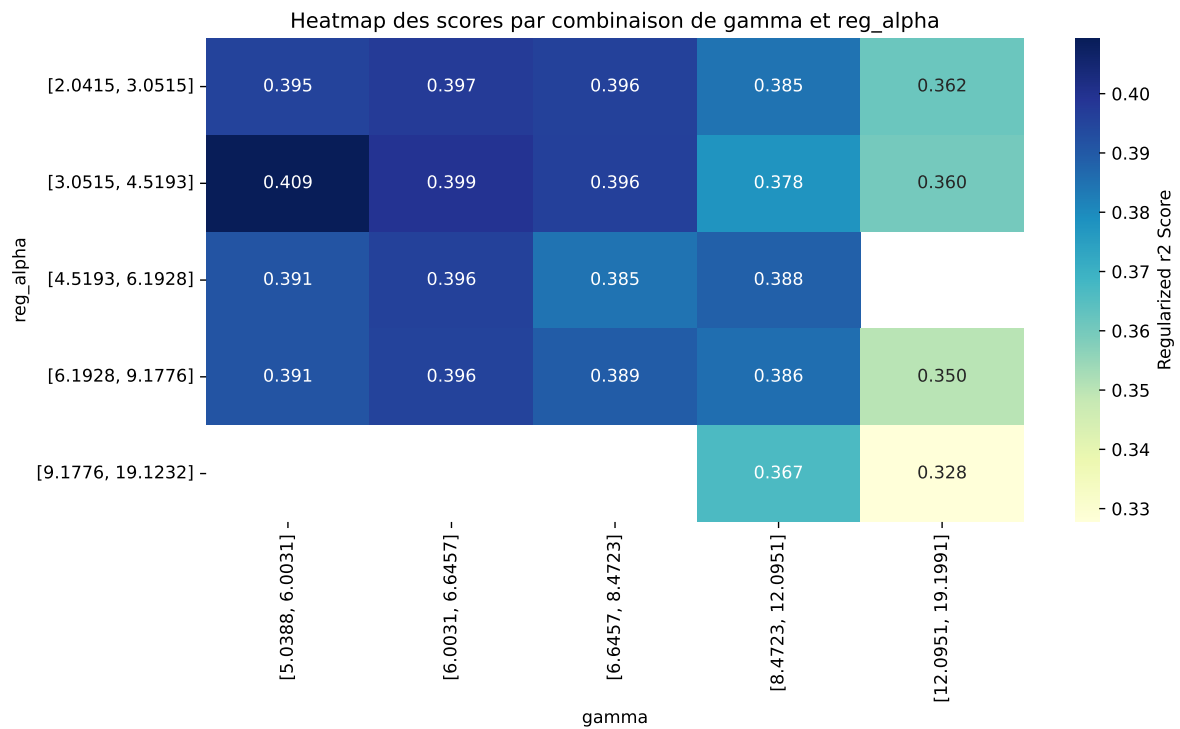
Paramètres avec >5% d'importance : ['gamma', 'reg_alpha', 'learning_rate']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4204	0.4104	0.4101	0.0331	0.4104	0.4288	0.0184	1.0448	1.0243
2	0.4193	0.4094	0.4084	0.0324	0.4094	0.4302	0.0209	1.051	1.0242
3	0.4187	0.4085	0.4075	0.0327	0.4085	0.4311	0.0226	1.0552	1.0249
4	0.4174	0.4078	0.4071	0.0334	0.4078	0.4259	0.0181	1.0443	1.0237
5	0.4156	0.4056	0.4044	0.0329	0.4056	0.4253	0.0197	1.0486	1.0248
6	0.4131	0.4041	0.4028	0.0337	0.4041	0.4273	0.0231	1.0573	1.0221
7	0.4127	0.4036	0.4023	0.033	0.4036	0.4206	0.017	1.0422	1.0226
8	0.4123	0.4033	0.4017	0.0331	0.4033	0.4238	0.0205	1.0509	1.0224
9	0.4092	0.4031	0.3988	0.0341	0.4031	0.4208	0.0177	1.0439	1.0152
10	0.409	0.4015	0.3982	0.0322	0.4015	0.4185	0.0169	1.0422	1.0187
---	---	---	---	---	---	---	---	---	---
70	0.308	0.3048	0.3029	0.0238	0.3048	0.3197	0.0148	1.0487	1.0102

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.012724861811927382, 'reg__subsample': 0.5428164568431036, 'reg__colsample_bytree': 0.6256893440077267, 'reg__colsample_bylevel': 0.5206693830309994, 'reg__colsample_bynode': 0.5727940291728868, 'reg__min_child_weight': 90, 'reg__gamma': 6.112548504770071, 'reg__reg_lambda': 10.462640298339153, 'reg__reg_alpha': 2.814122194567575, 'reg__max_delta_step': 2}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.011755787442566947, 'reg__subsample': 0.5440137376811882, 'reg__colsample_bytree': 0.49011497903816487, 'reg__colsample_bylevel': 0.5049279089725307, 'reg__colsample_bynode': 0.5422999635930567, 'reg__min_child_weight': 80, 'reg__gamma': 5.591771737849315, 'reg__reg_lambda': 10.194198077295964, 'reg__reg_alpha': 3.4909904210953053, 'reg__max_delta_step': 2}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.014260087596625455, 'reg__subsample': 0.6096062754042412, 'reg__colsample_bytree': 0.4089293629634827, 'reg__colsample_bylevel': 0.4935984374493647, 'reg__colsample_bynode': 0.5543709902545464, 'reg__min_child_weight': 100, 'reg__gamma': 5.967371674693401, 'reg__reg_lambda': 16.540656884174226, 'reg__reg_alpha': 2.0990854044074814, 'reg__max_delta_step': 2}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.012055596739281652, 'reg__subsample': 0.547162474779377, 'reg__colsample_bytree': 0.625402557385135, 'reg__colsample_bylevel': 0.5152893637561747, 'reg__colsample_bynode': 0.5742674213875265, 'reg__min_child_weight': 90, 'reg__gamma': 6.126355004093982, 'reg__reg_lambda': 12.43737308288648, 'reg__reg_alpha': 3.06046811354203, 'reg__max_delta_step': 2}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.014088239599715339, 'reg__subsample': 0.6102034237082845, 'reg__colsample_bytree': 0.524867325184658, 'reg__colsample_bylevel': 0.5171368339661655, 'reg__colsample_bynode': 0.5573350425833039, 'reg__min_child_weight': 100, 'reg__gamma': 6.2089820147510615, 'reg__reg_lambda': 16.353834518443612, 'reg__reg_alpha': 3.436332252068574, 'reg__max_delta_step': 2}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.01245754875004913, 'reg__subsample': 0.5441833728065562, 'reg__colsample_bytree': 0.6095533486647111, 'reg__colsample_bylevel': 0.6451004355915584, 'reg__colsample_bynode': 0.5713345975950058, 'reg__min_child_weight': 90, 'reg__gamma': 6.500417668565499, 'reg__reg_lambda': 24.87908486645527, 'reg__reg_alpha': 2.7023278586242334, 'reg__max_delta_step': 1}

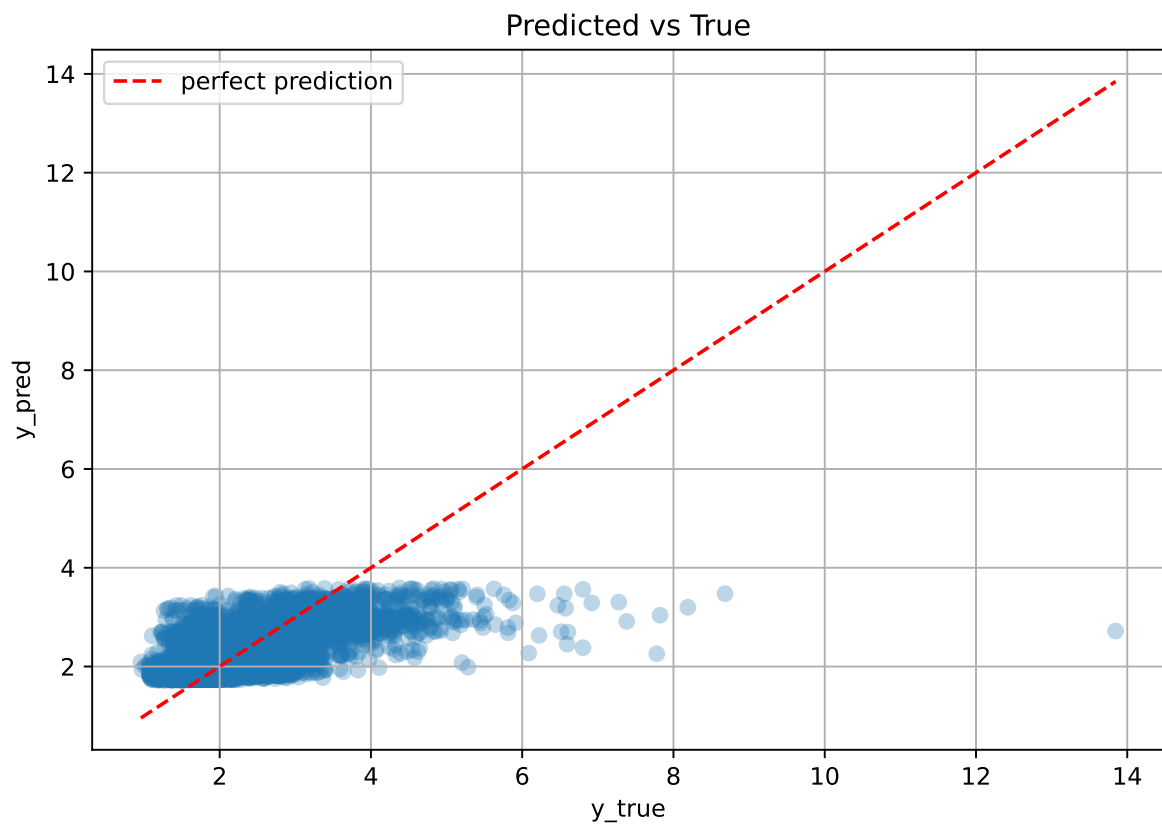
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.011849197297434026, 'reg__subsample': 0.5794810607508005, 'reg__colsample_bytree': 0.6272581396472058, 'reg__colsample_bylevel': 0.5325194834247221, 'reg__colsample_bynode': 0.552522603121262, 'reg__min_child_weight': 90, 'reg__gamma': 6.449419088010762, 'reg__reg_lambda': 18.140140487174705, 'reg__reg_alpha': 2.832891658001109, 'reg__max_delta_step': 2}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.013948543914044759, 'reg__subsample': 0.5483569993989302, 'reg__colsample_bytree': 0.49259463997510783, 'reg__colsample_bylevel': 0.6423788845778364, 'reg__colsample_bynode': 0.5838983298008111, 'reg__min_child_weight': 90, 'reg__gamma': 6.225806084166909, 'reg__reg_lambda': 26.610920416586772, 'reg__reg_alpha': 3.144230713845399, 'reg__max_delta_step': 1}

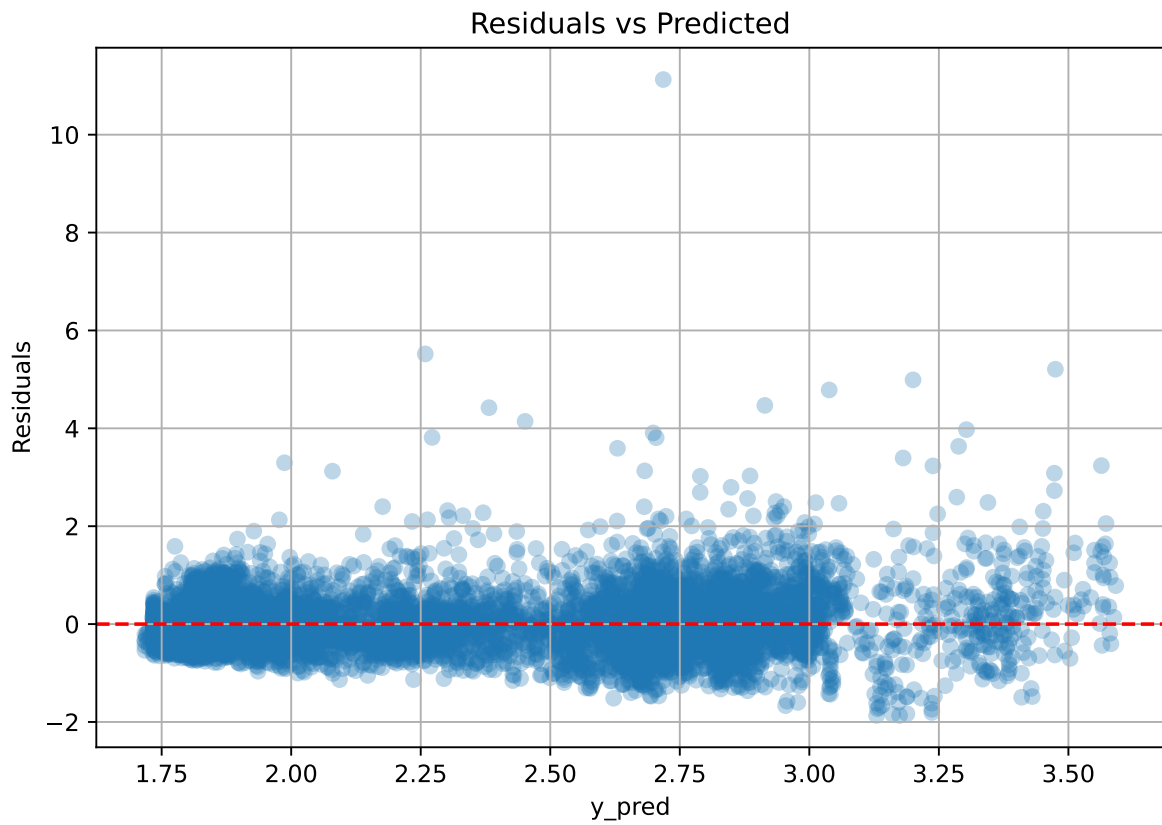
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.015744608493450694, 'reg__subsample': 0.5668265163657821, 'reg__colsample_bytree': 0.6579717861860632, 'reg__colsample_bylevel': 0.5586645641797348, 'reg__colsample_bynode': 0.5245359575776769, 'reg__min_child_weight': 80, 'reg__gamma': 8.53070296442054, 'reg__reg_lambda': 10.506004085249531, 'reg__reg_alpha': 4.7669649447409235, 'reg__max_delta_step': 3}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.01550887813896165, 'reg__subsample': 0.5678041989567782, 'reg__colsample_bytree': 0.590504620405936, 'reg__colsample_bylevel': 0.5593364925997144, 'reg__colsample_bynode': 0.5243049492345256, 'reg__min_child_weight': 80, 'reg__gamma': 8.070235606862576, 'reg__reg_lambda': 10.921577176578579, 'reg__reg_alpha': 4.6490129360550005, 'reg__max_delta_step': 3}

2.4.2 Scatter



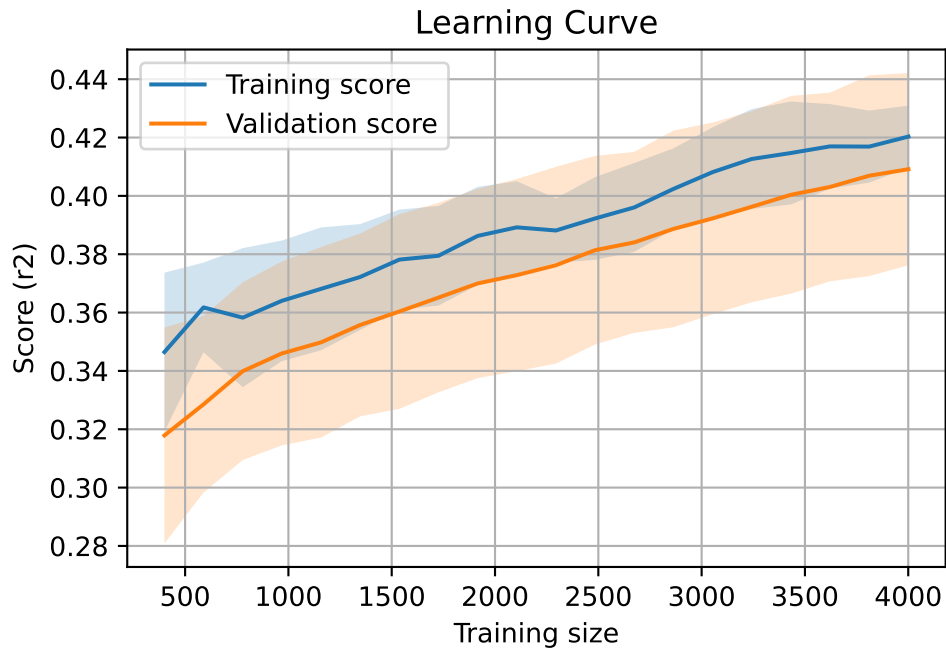
2.4.3 Scatter residuals



2.4.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450

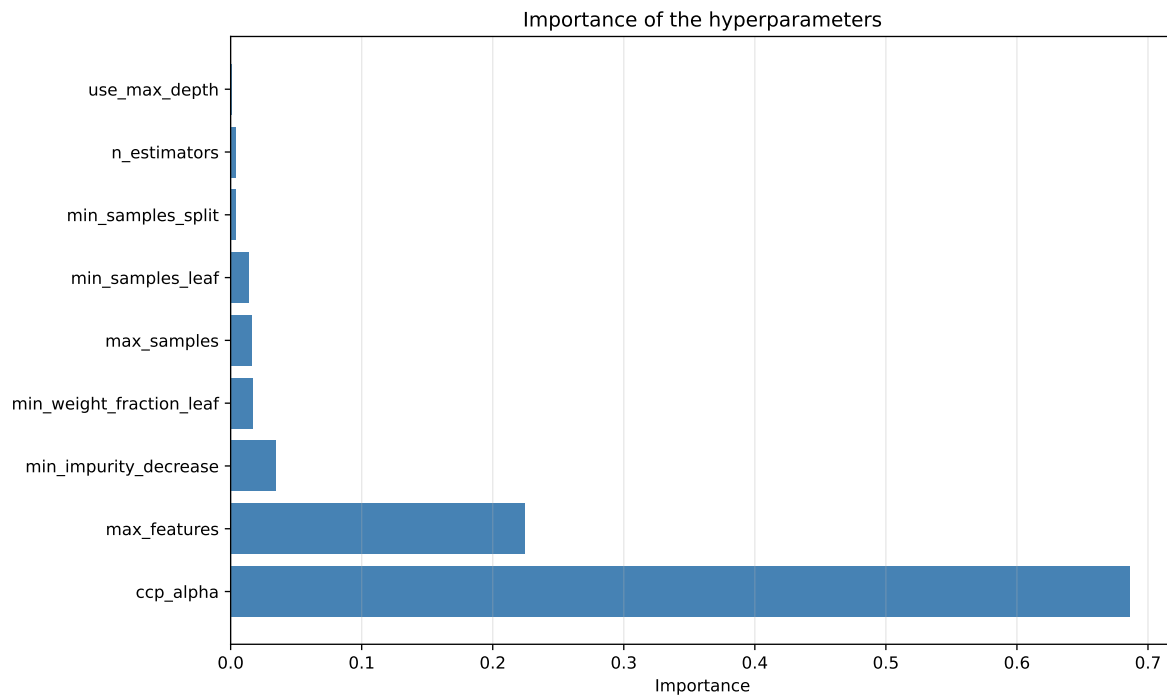
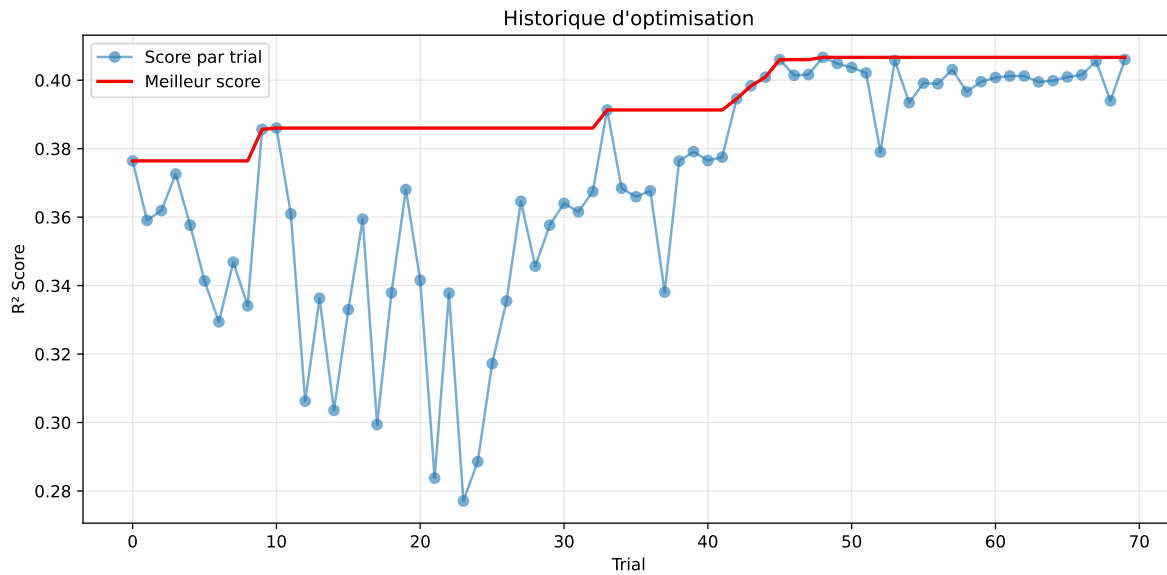


2.5 Model : RFR

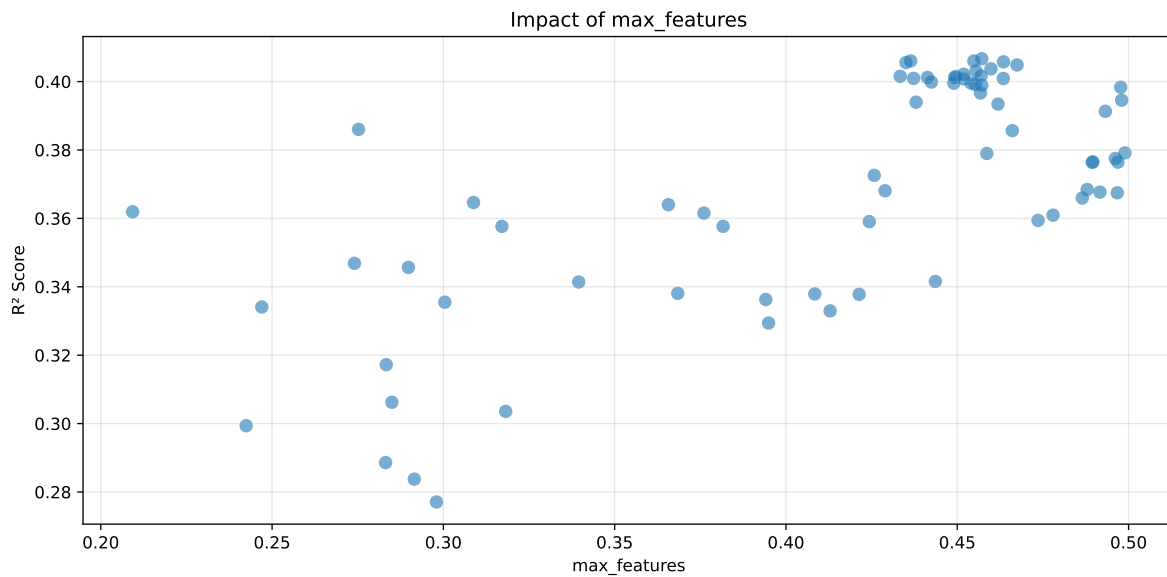
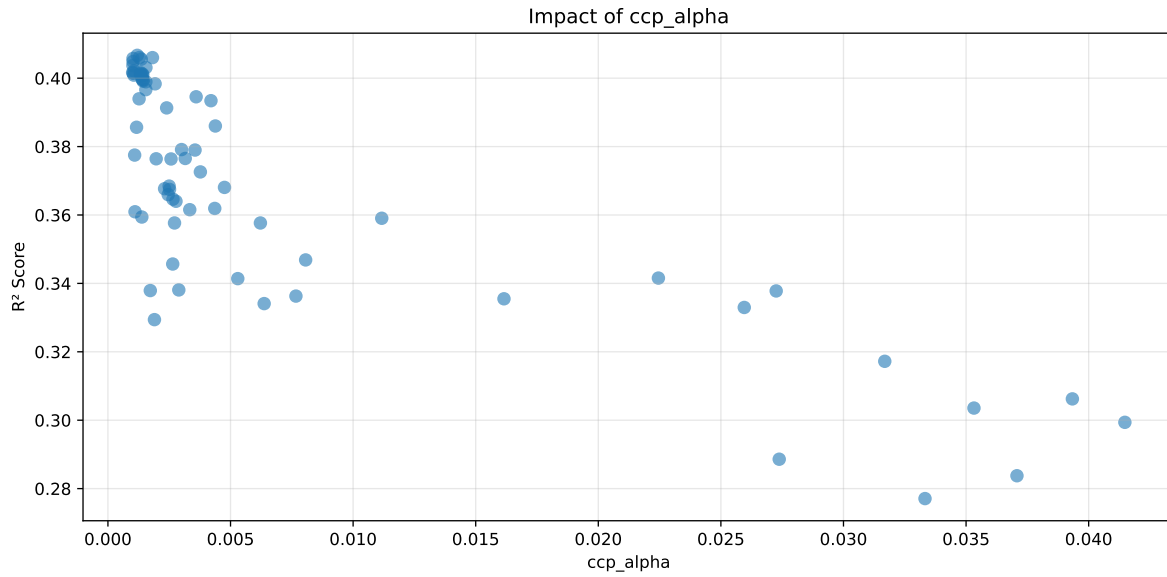
2.5.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845
y_test: mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====
feature importance r_umidity 0.513233 r_temperature 0.254863 r_THI 0.153687 r_NH3
0.046304 r_CO2 0.031911

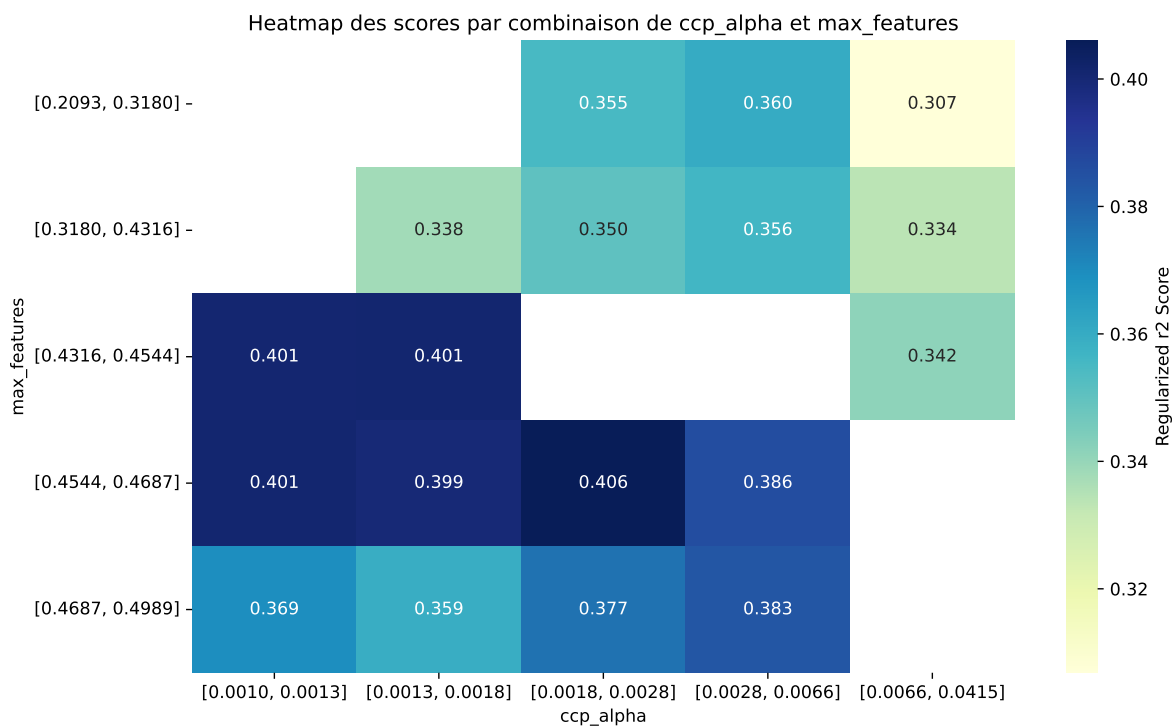


Paramètres avec >5% d'importance : ['ccp_alpha', 'max_features']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4174	0.4067	0.4055	0.0346	0.4067	0.4058	-0.0009	0.9978	1.0264
2	0.4173	0.406	0.4055	0.0344	0.406	0.4059	-0.0001	0.9997	1.0277
3	0.4165	0.406	0.4047	0.0344	0.406	0.4044	-0.0016	0.9959	1.0259
4	0.4162	0.4058	0.4045	0.0345	0.4058	0.4047	-0.0011	0.9973	1.0257
5	0.4168	0.4056	0.405	0.0345	0.4056	0.4053	-0.0002	0.9995	1.0277
6	0.4152	0.4049	0.4034	0.0345	0.4049	0.4038	-0.0011	0.9974	1.0256
7	0.4137	0.4037	0.4018	0.0345	0.4037	0.4016	-0.0021	0.9948	1.0247
8	0.4132	0.4031	0.4011	0.0345	0.4031	0.4019	-0.0012	0.9969	1.0249
9	0.4119	0.4021	0.3999	0.0342	0.4021	0.3996	-0.0025	0.9938	1.0242
10	0.4112	0.4017	0.3993	0.0341	0.4017	0.3993	-0.0023	0.9942	1.0239
---	---	---	---	---	---	---	---	---	---
70	0.2788	0.2771	0.276	0.0227	0.2771	0.2782	0.0012	1.0042	1.0062

Hyperparameters:

Rank 1: {'reg__n_estimators': 450, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 14, 'reg__max_features': 0.4571634234088179, 'reg__max_samples': 0.7176913510011629, 'reg__min_impurity_decrease': 0.0020100774208573246, 'reg__ccp_alpha': 0.0011967669808998988, 'reg__min_weight_fraction_leaf': 0.0542245815156927}

Rank 2: {'reg__n_estimators': 500, 'reg__min_samples_split': 40, 'reg__min_samples_leaf': 6, 'reg__max_features': 0.43640531109063946, 'reg__max_samples': 0.7419816399006193, 'reg__min_impurity_decrease': 0.002405771053390482, 'reg__ccp_alpha': 0.0012837410470768157, 'reg__min_weight_fraction_leaf': 0.05102957023318214}

Rank 3: {'reg__n_estimators': 450, 'reg__min_samples_split': 35, 'reg__min_samples_leaf': 10, 'reg__max_features': 0.454853259548969, 'reg__max_samples': 0.6636139746326648, 'reg__min_impurity_decrease': 0.002039770639697269, 'reg__ccp_alpha': 0.0018160627684686338, 'reg__min_weight_fraction_leaf': 0.05457237601110866}

Rank 4: {'reg__n_estimators': 450, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 8, 'reg__max_features': 0.4634955144780261, 'reg__max_samples': 0.6546108940348051, 'reg__min_impurity_decrease': 0.002069789869421707, 'reg__ccp_alpha': 0.00102178648715604, 'reg__min_weight_fraction_leaf': 0.054892484089696436}

Rank 5: {'reg__n_estimators': 500, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 14, 'reg__max_features': 0.43503127456357643, 'reg__max_samples': 0.7434726551434634, 'reg__min_impurity_decrease': 0.0023474732929423215, 'reg__ccp_alpha': 0.0013585881048546057, 'reg__min_weight_fraction_leaf': 0.05191018393686362}

Rank 6: {'reg__n_estimators': 450, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 14, 'reg__max_features': 0.46743469609763844, 'reg__max_samples': 0.7193908452068944, 'reg__min_impurity_decrease': 0.0021914225550160624, 'reg__ccp_alpha': 0.0010230837306244413, 'reg__min_weight_fraction_leaf': 0.054657460499272253}

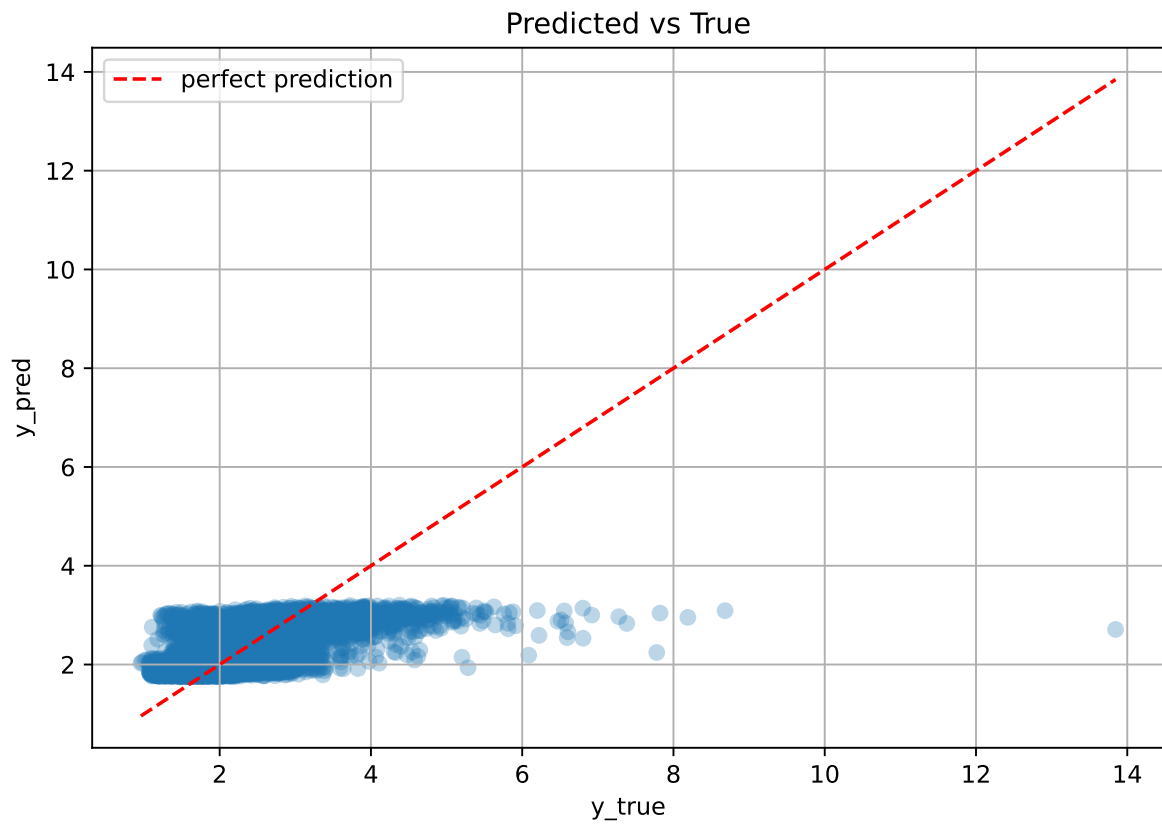
Rank 7: {'reg__n_estimators': 450, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 14, 'reg__max_features': 0.45983927710482625, 'reg__max_samples': 0.7313285434368809, 'reg__min_impurity_decrease': 0.002043353065023669, 'reg__ccp_alpha': 0.0010181206397401853, 'reg__min_weight_fraction_leaf': 0.05726925688011359}

Rank 8: {'reg__n_estimators': 450, 'reg__min_samples_split': 15, 'reg__min_samples_leaf': 8, 'reg__max_features': 0.4554411814688921, 'reg__max_samples': 0.721810473747634, 'reg__min_impurity_decrease': 0.0022336653974923187, 'reg__ccp_alpha': 0.0015501102458493686, 'reg__min_weight_fraction_leaf': 0.05642375757675747}

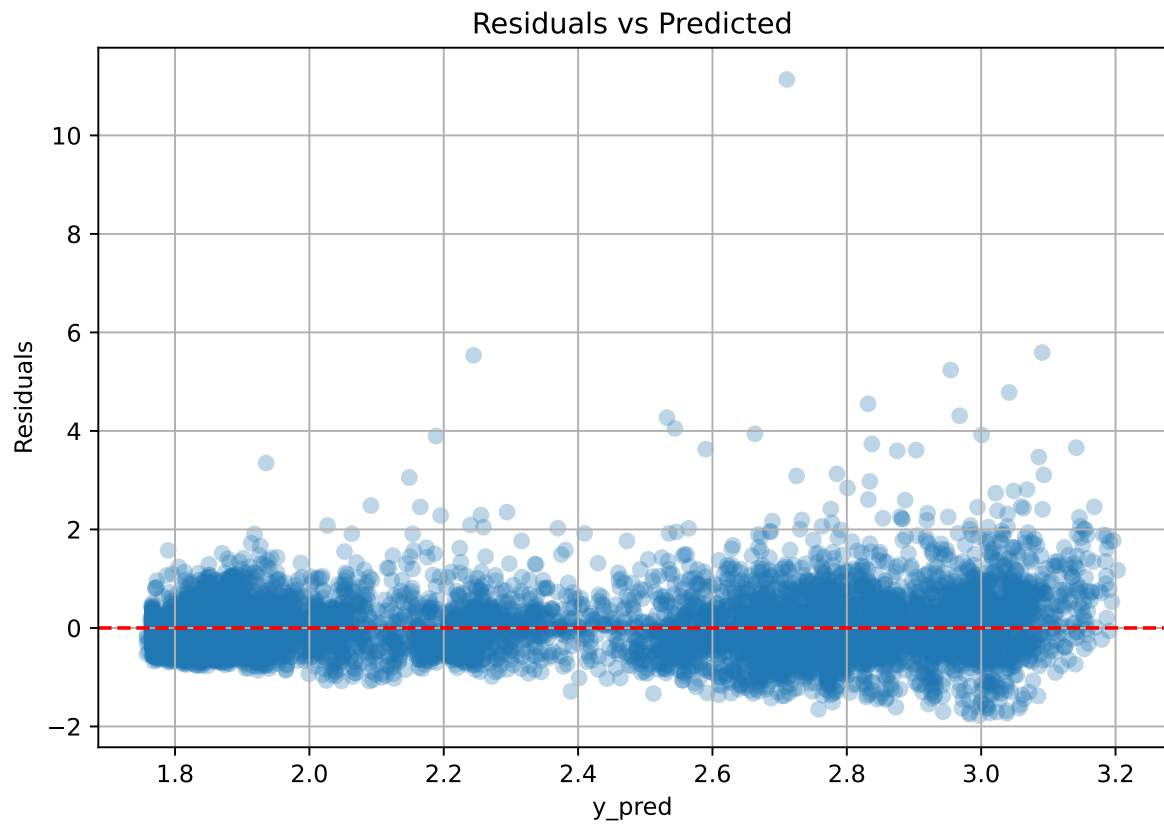
Rank 9: {'reg__n_estimators': 450, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 14, 'reg__max_features': 0.4518492227097892, 'reg__max_samples': 0.7212996409845812, 'reg__min_impurity_decrease': 0.0020508130116035205, 'reg__ccp_alpha': 0.001069855231560266, 'reg__min_weight_fraction_leaf': 0.0587725993037866}

Rank 10: {'reg__n_estimators': 450, 'reg__min_samples_split': 45, 'reg__min_samples_leaf': 14, 'reg__max_features': 0.4569531847358641, 'reg__max_samples': 0.7129351845644158, 'reg__min_impurity_decrease': 0.002193764305305184, 'reg__ccp_alpha': 0.0010075211029610671, 'reg__min_weight_fraction_leaf': 0.05847720917534821}

2.5.2 Scatter



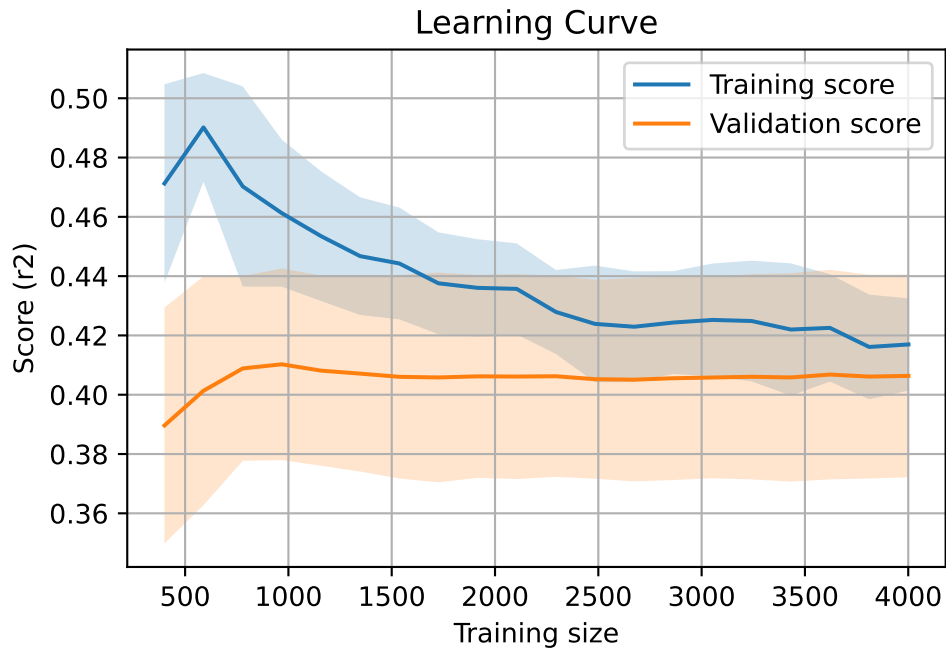
2.5.3 Scatter residuals



2.5.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train	penalized_score
Linear	0.36	0.365	0.037	0.3451	0.3453	-0.02	0.946	0.986	nan
XGB	0.42	0.41	0.033	0.4101	0.4288	0.018	1.045	1.024	0.41
RFR	0.417	0.407	0.035	0.4055	0.4058	-0.001	0.998	1.026	0.407

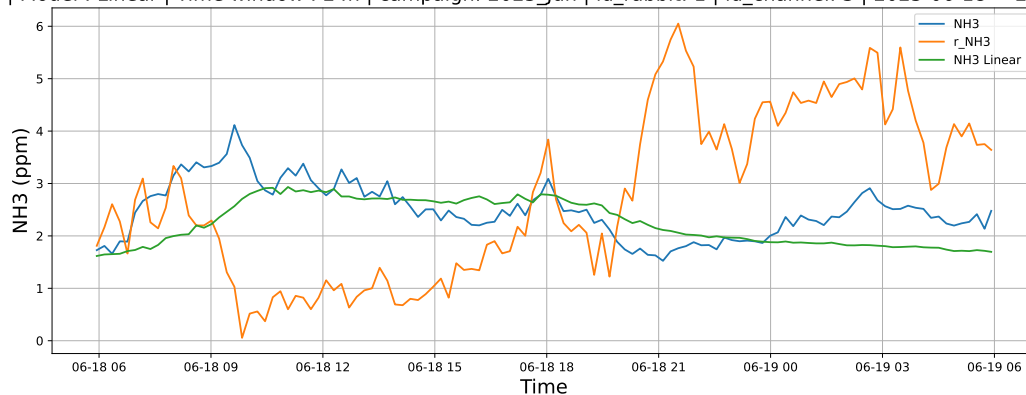
3 Plot

3.1 Standalone plots

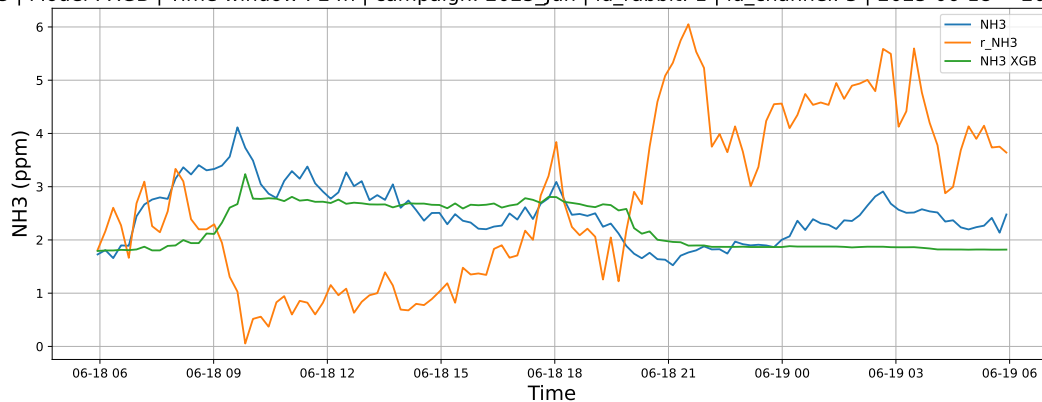
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

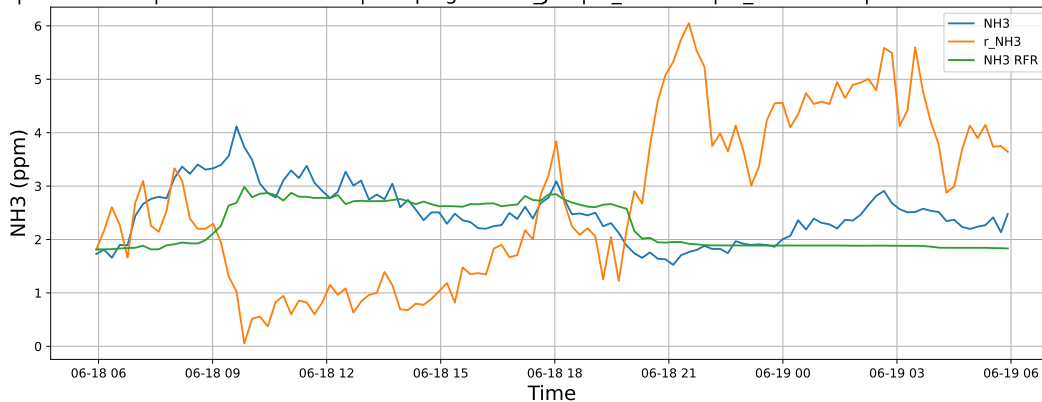
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : RFR | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

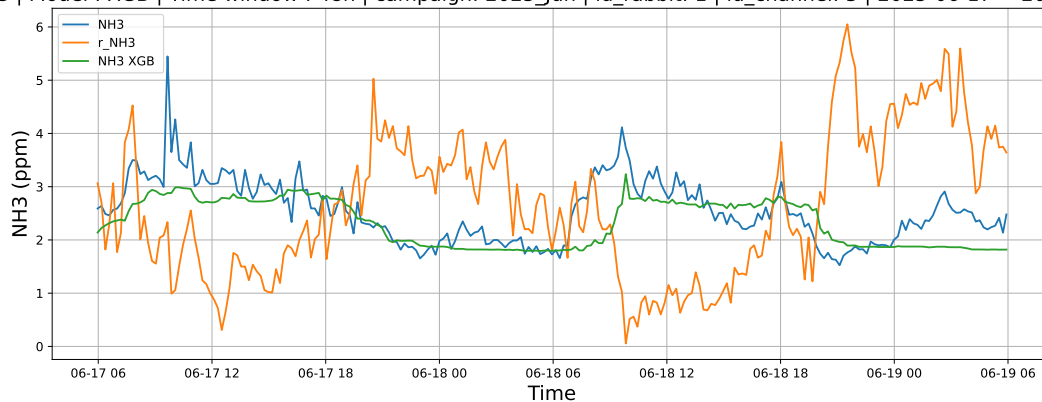


3.1.1.2 Time window : 48

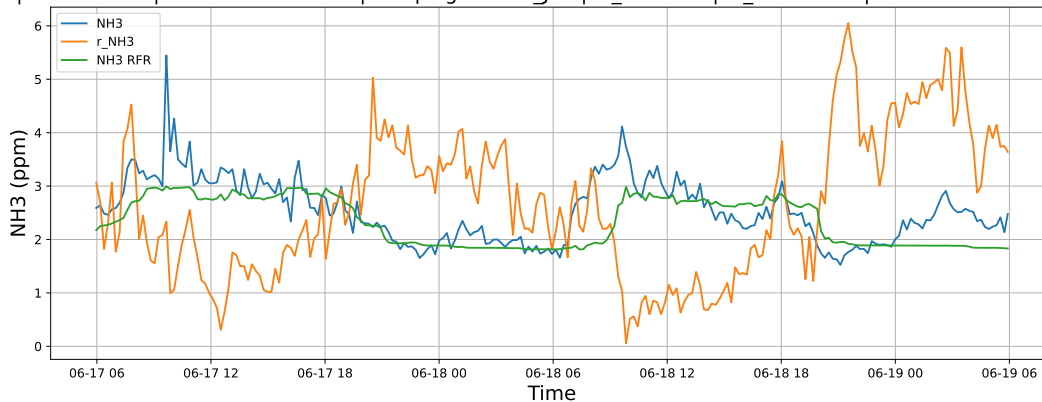
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

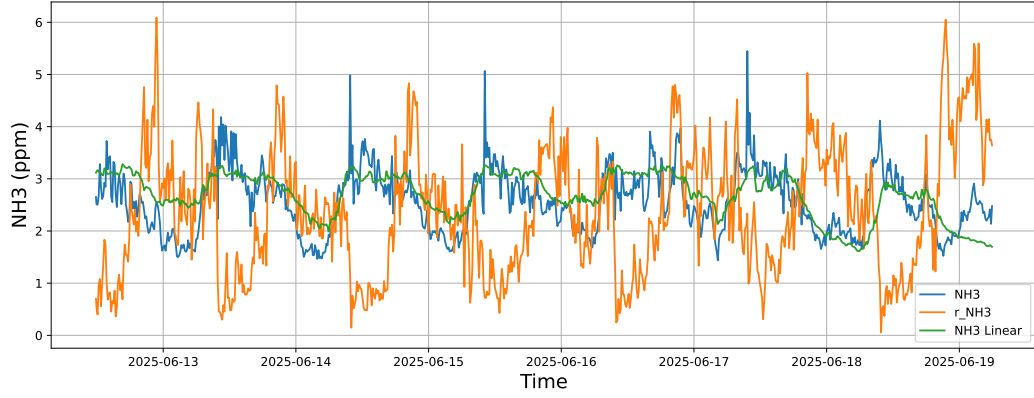


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

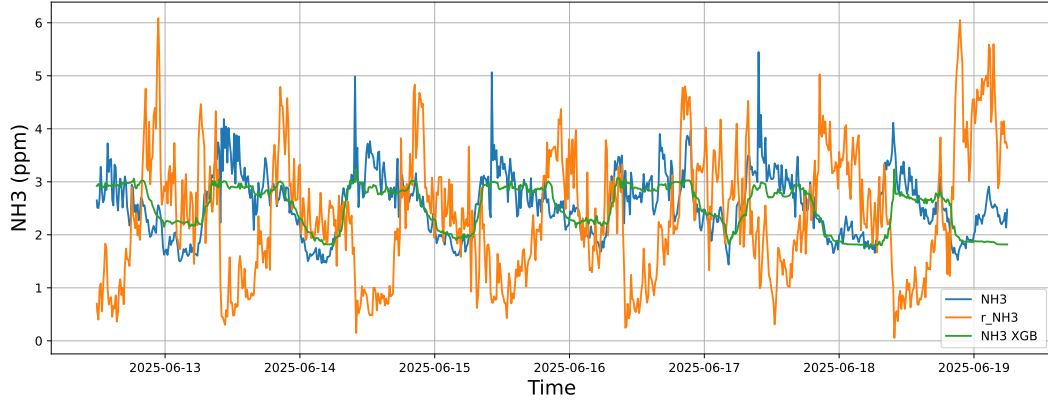


3.1.1.3 Time window : ALL

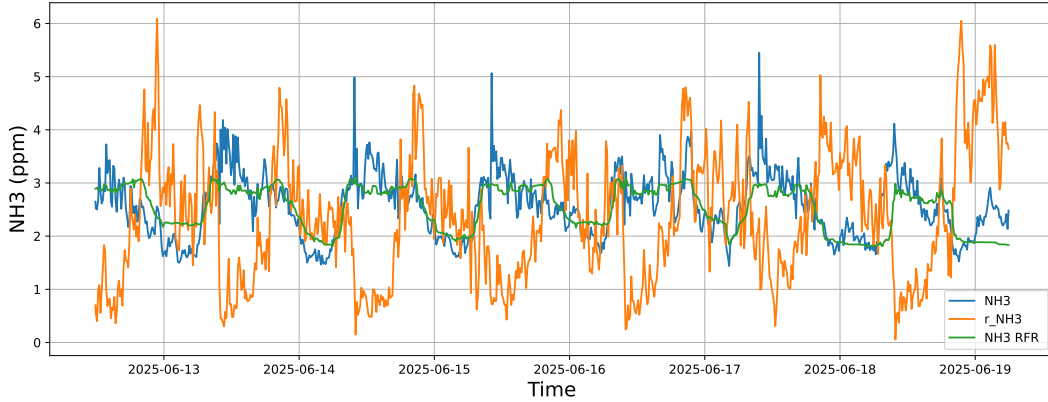
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



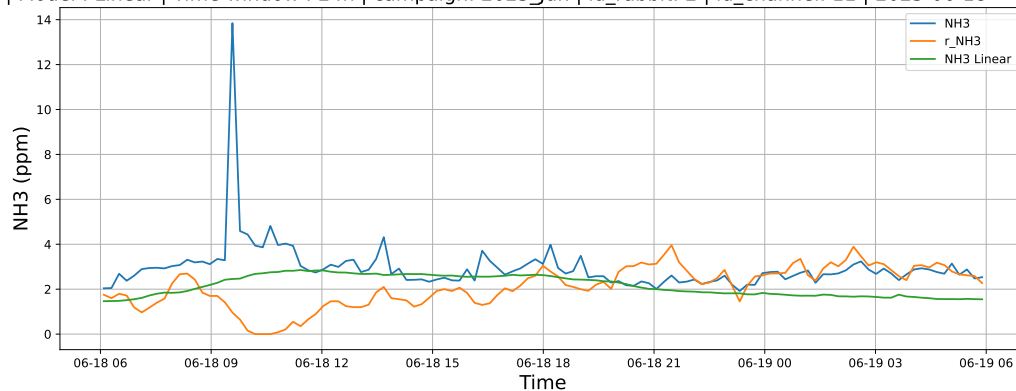
NH3 | Model : RFR | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



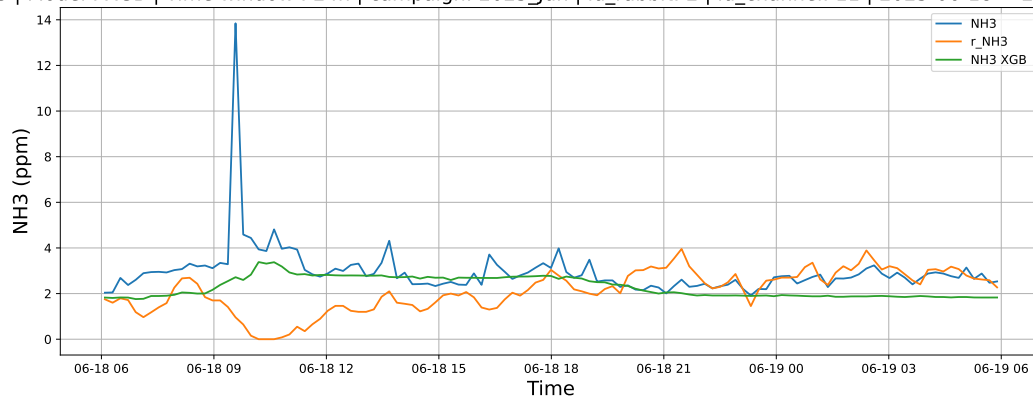
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

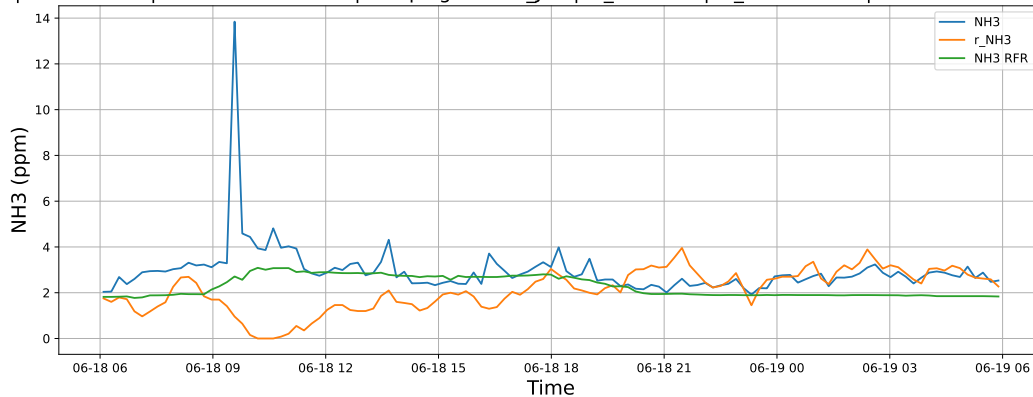
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

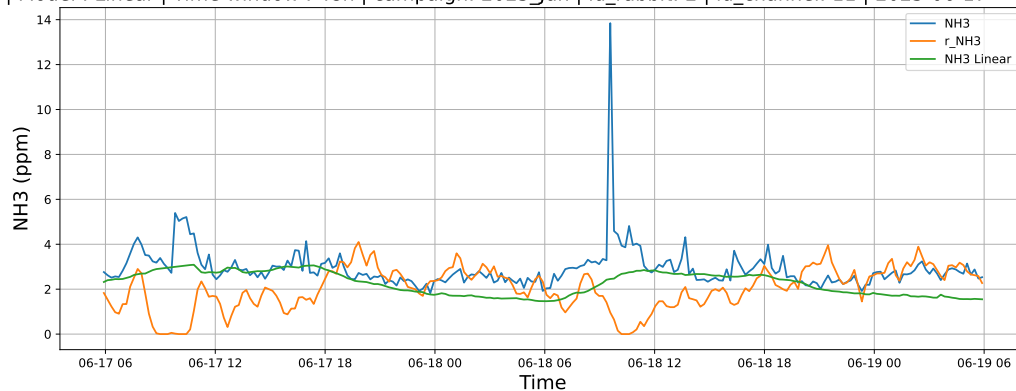


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

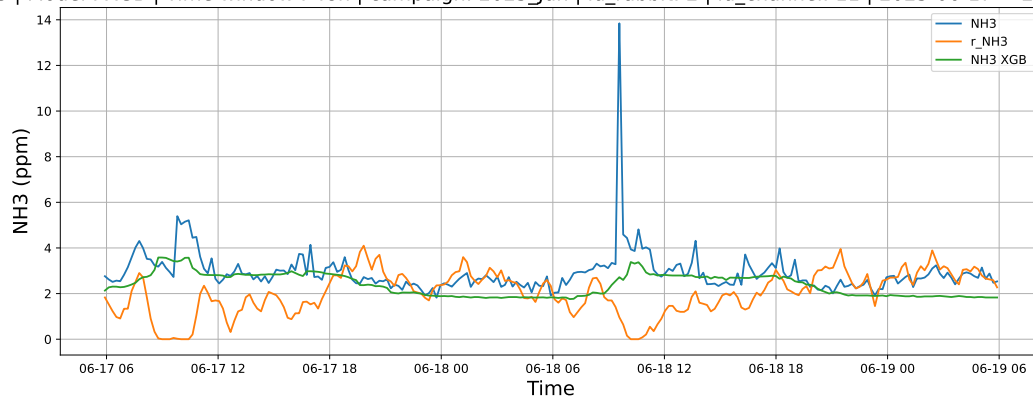


3.1.2.2 Time window : 48

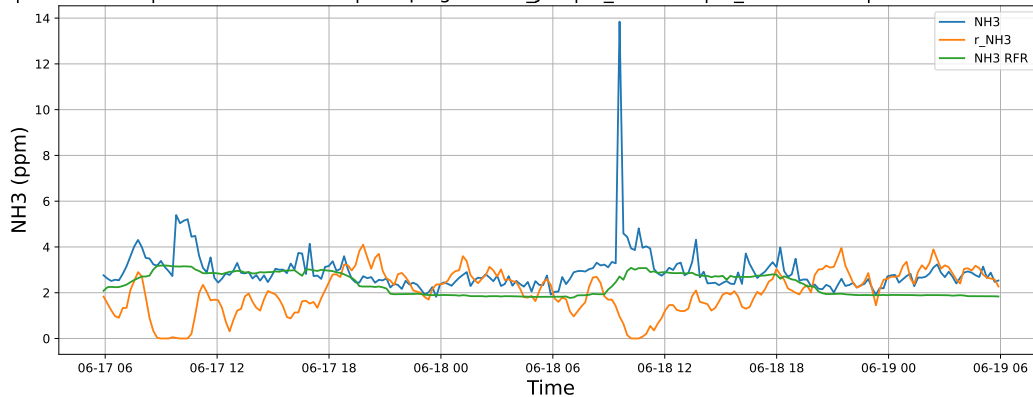
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

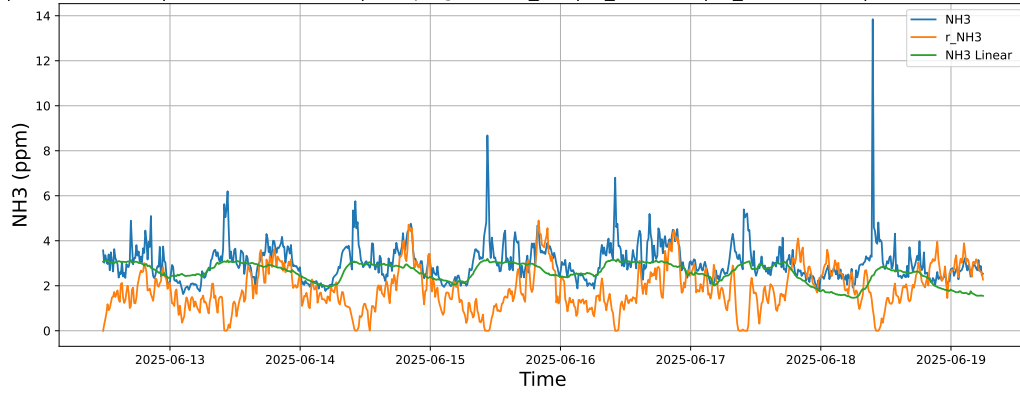


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

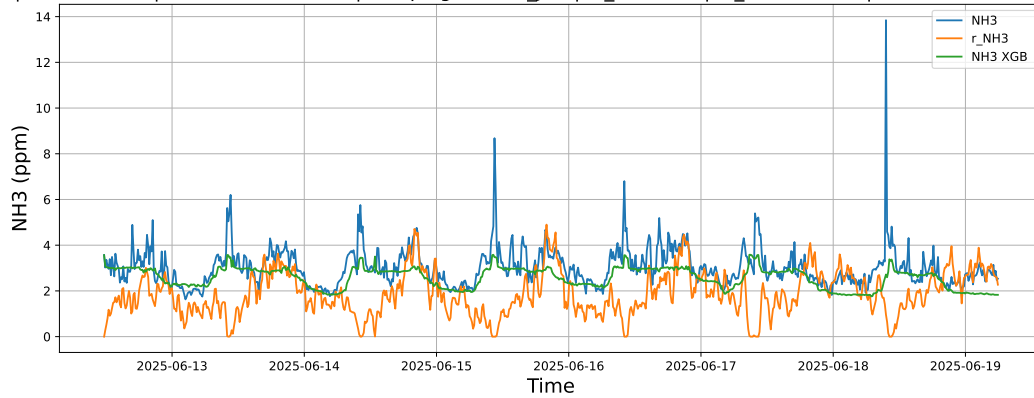


3.1.2.3 Time window : ALL

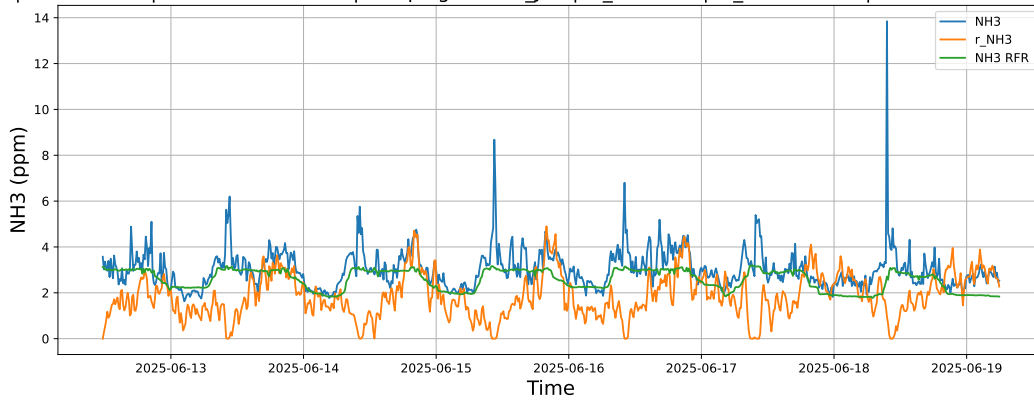
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



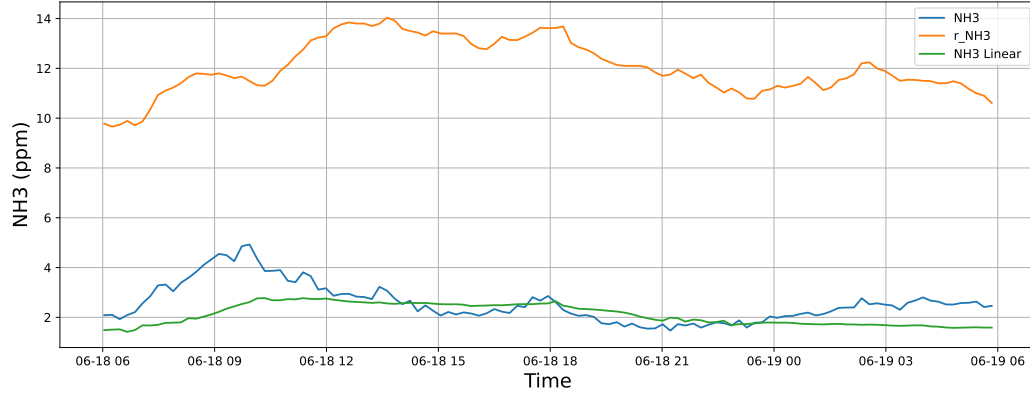
NH3 | Model : RFR | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



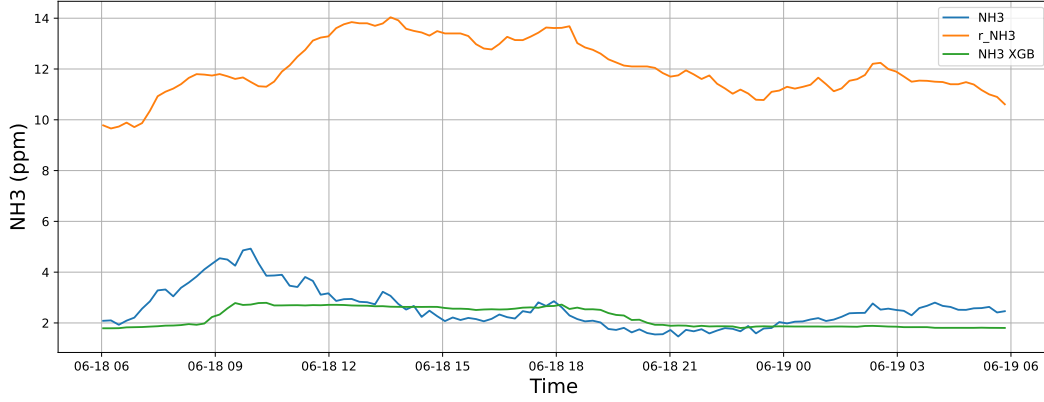
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

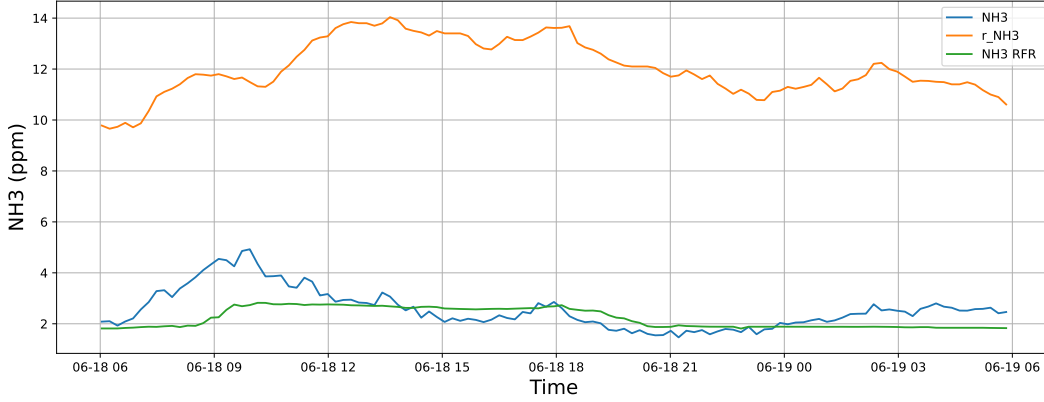
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

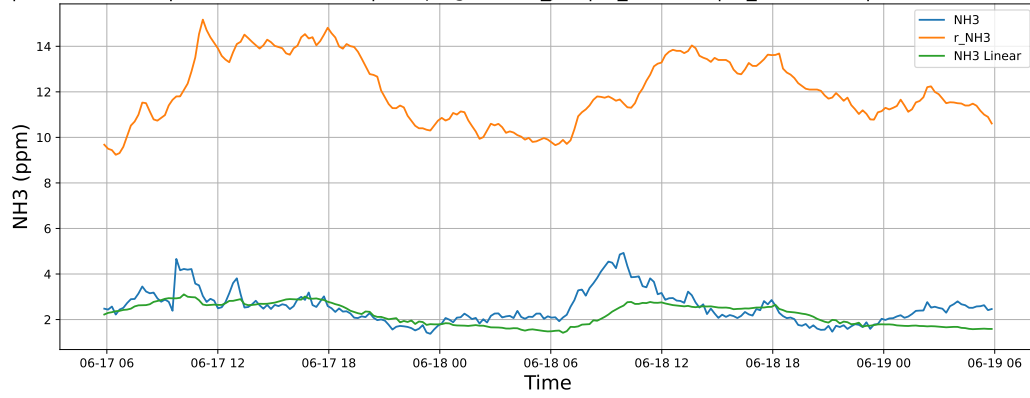


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

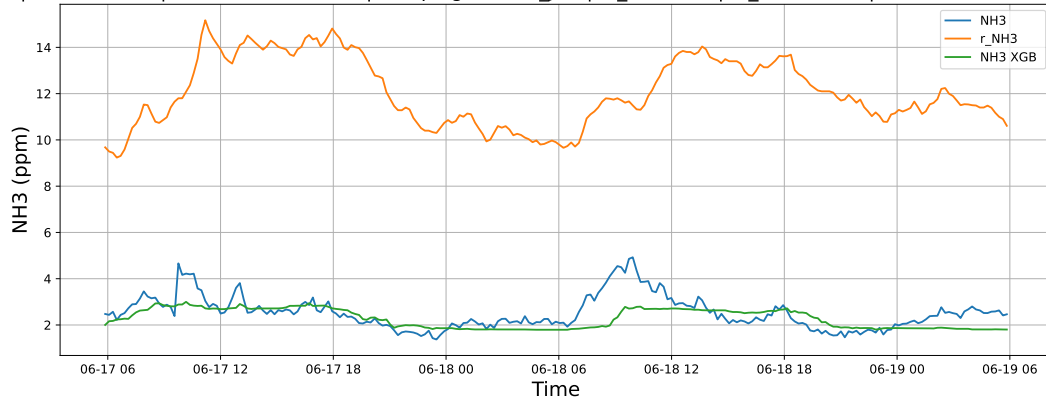


3.1.3.2 Time window : 48

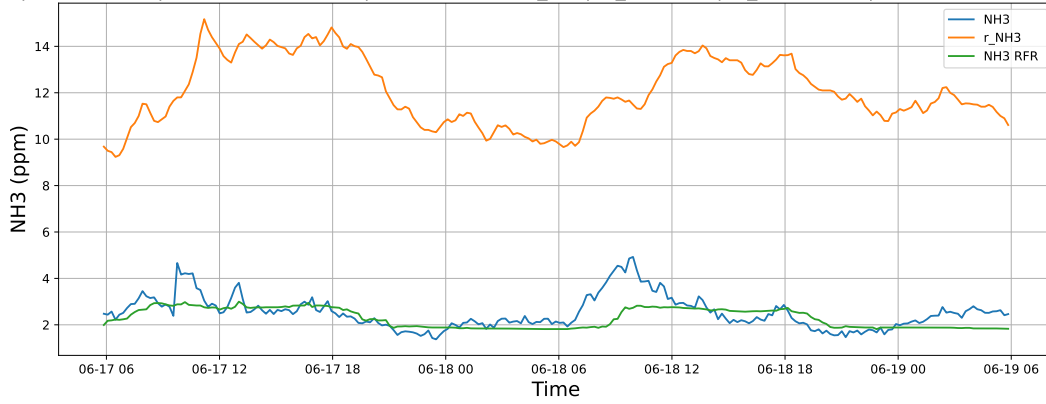
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

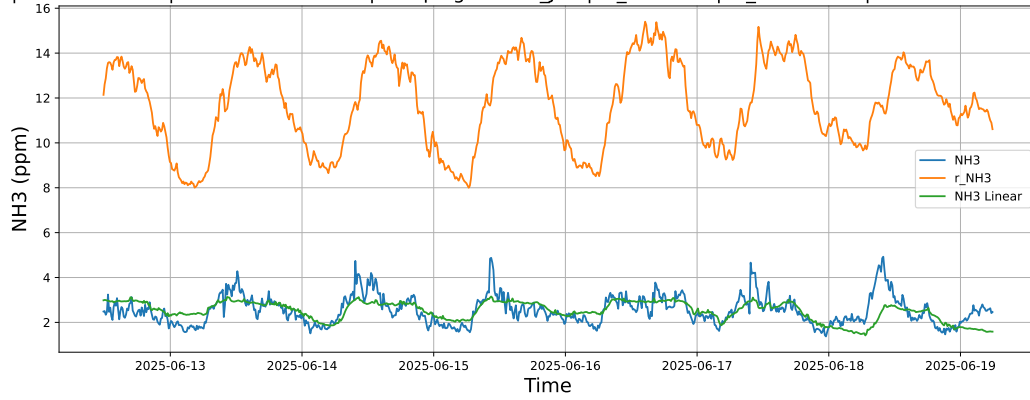


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

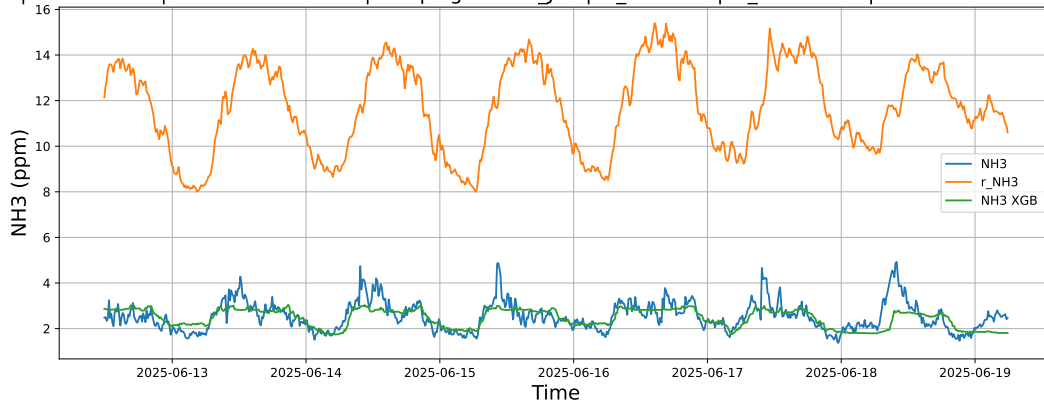


3.1.3.3 Time window : ALL

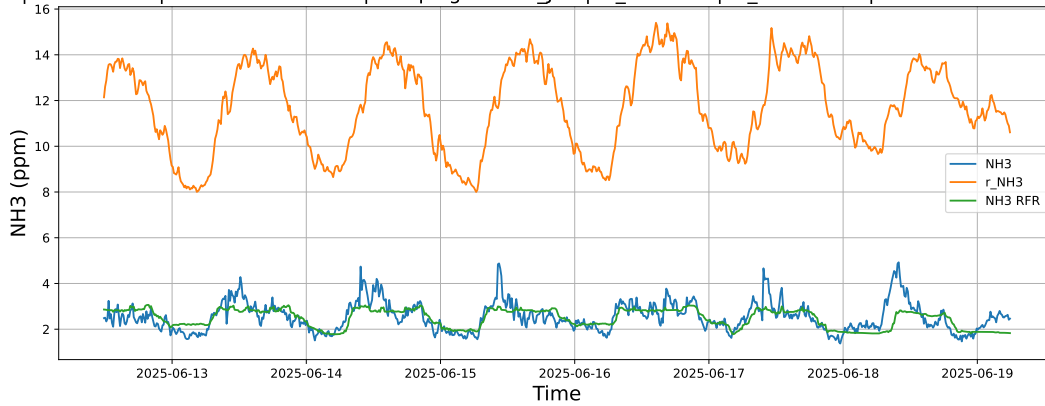
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : RFR | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



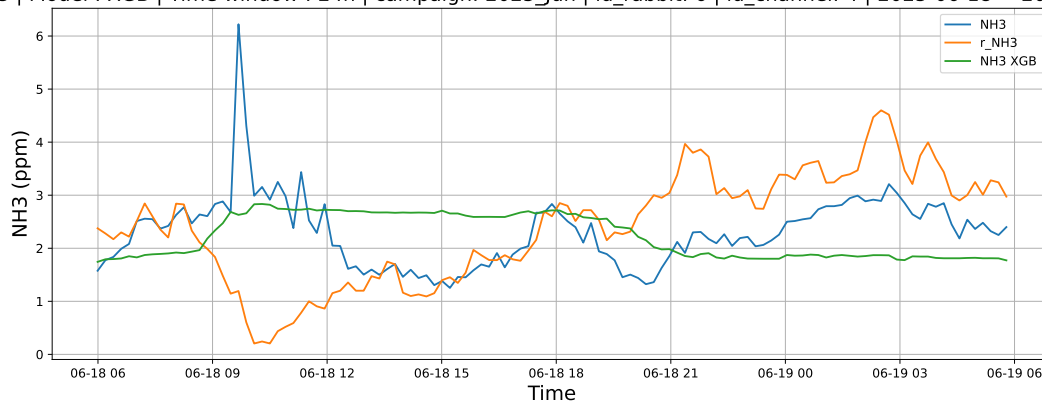
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

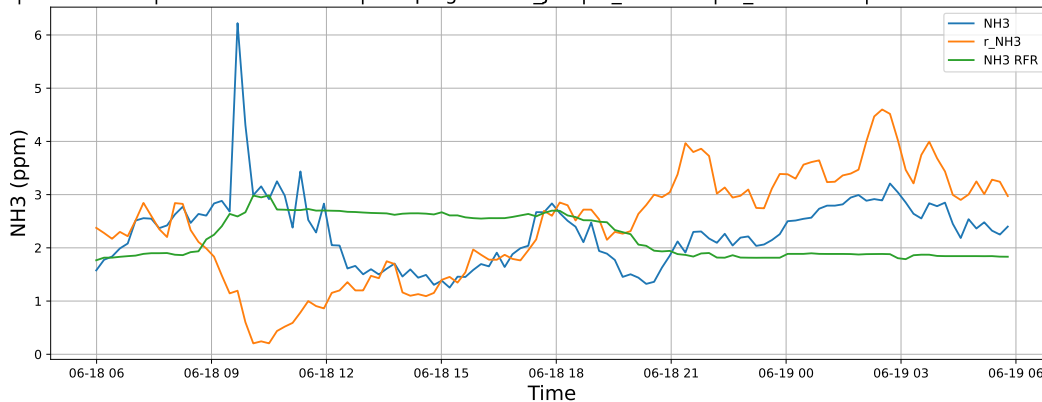
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

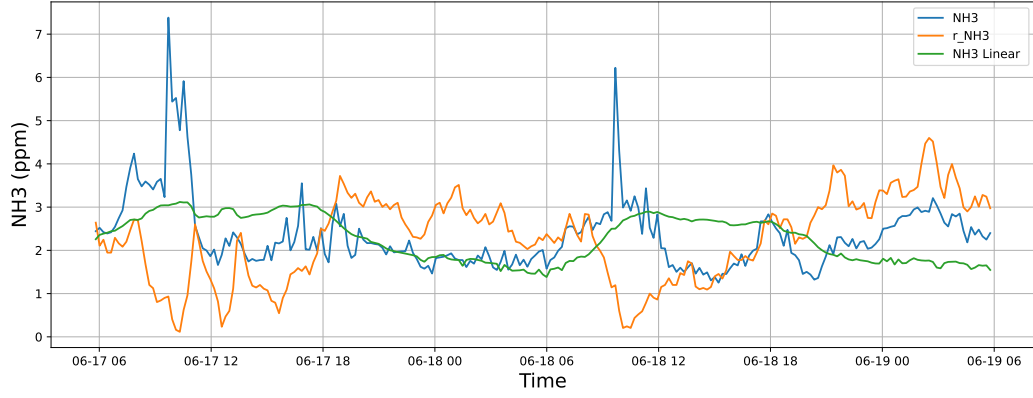


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

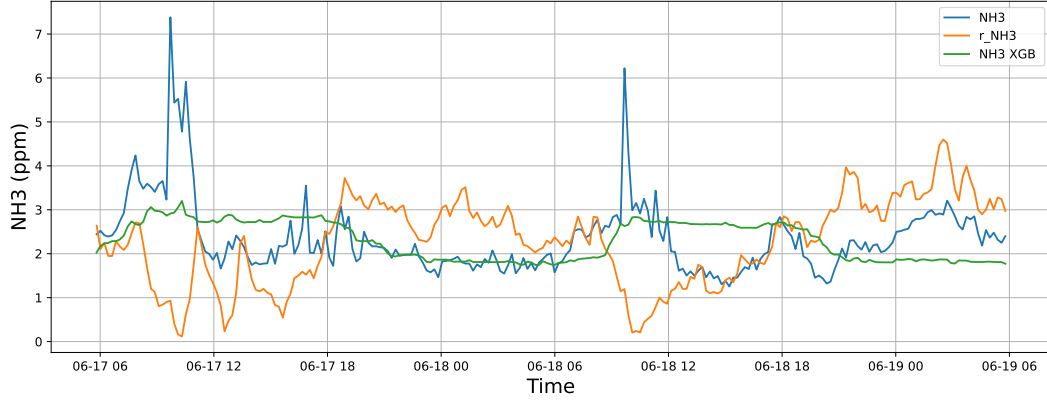


3.1.4.2 Time window : 48

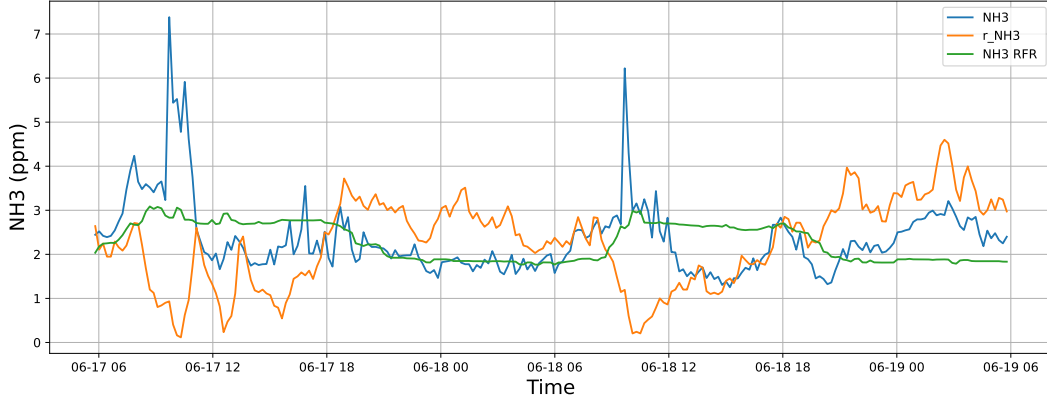
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

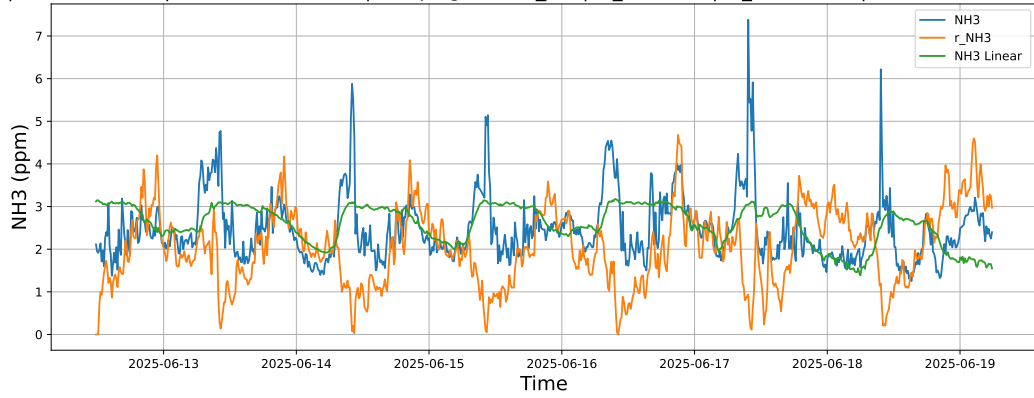


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

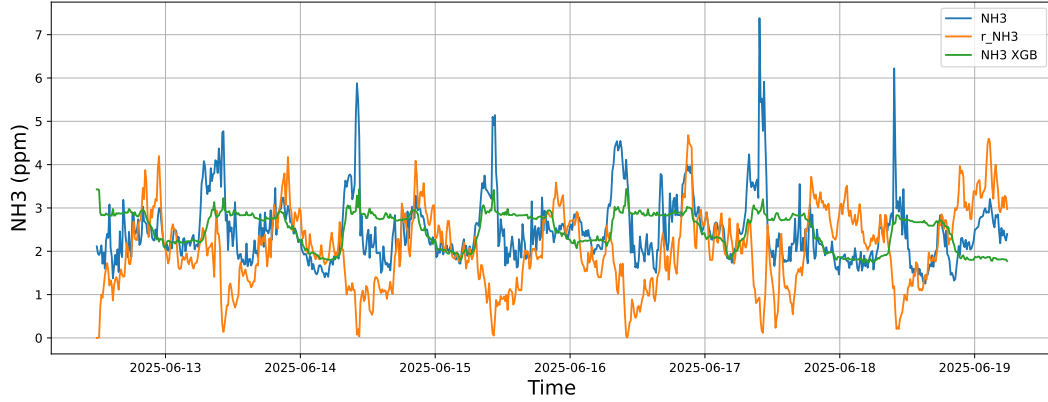


3.1.4.3 Time window : ALL

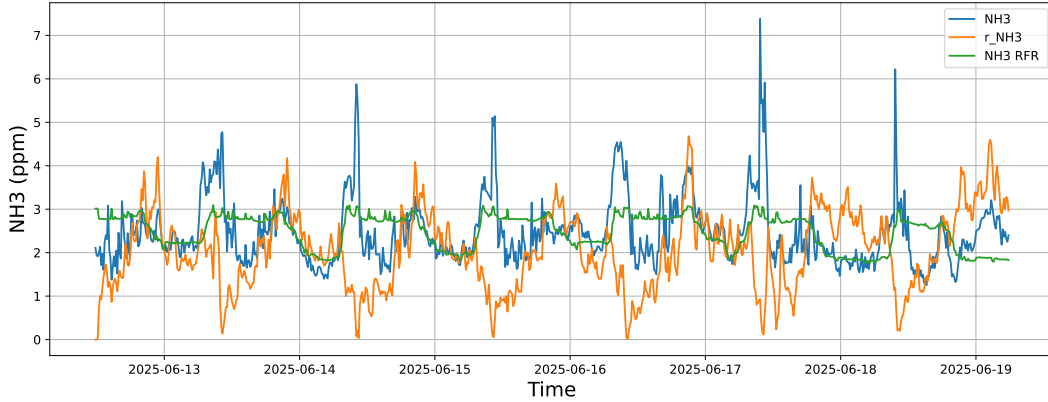
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



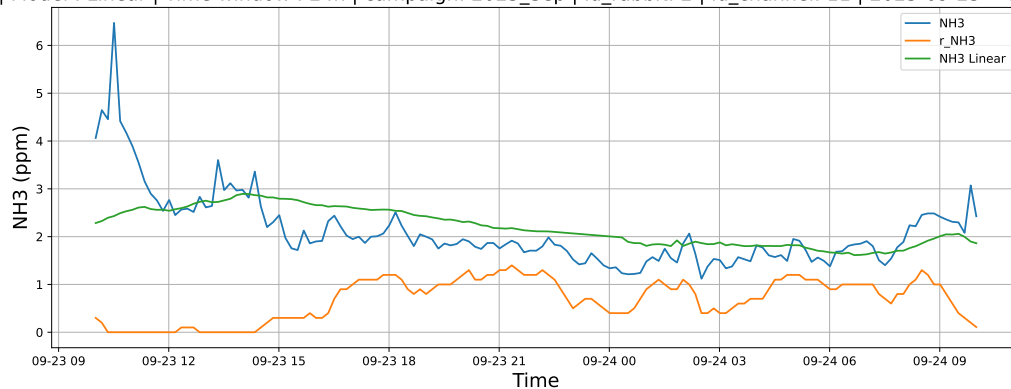
NH3 | Model : RFR | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



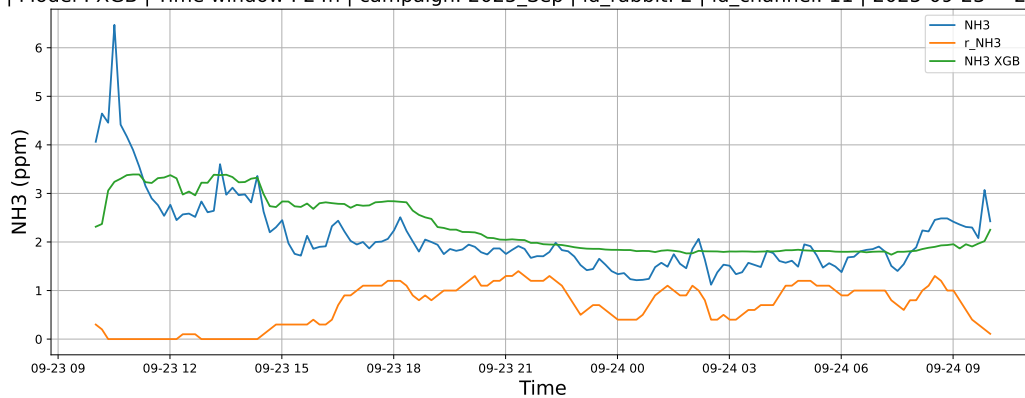
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

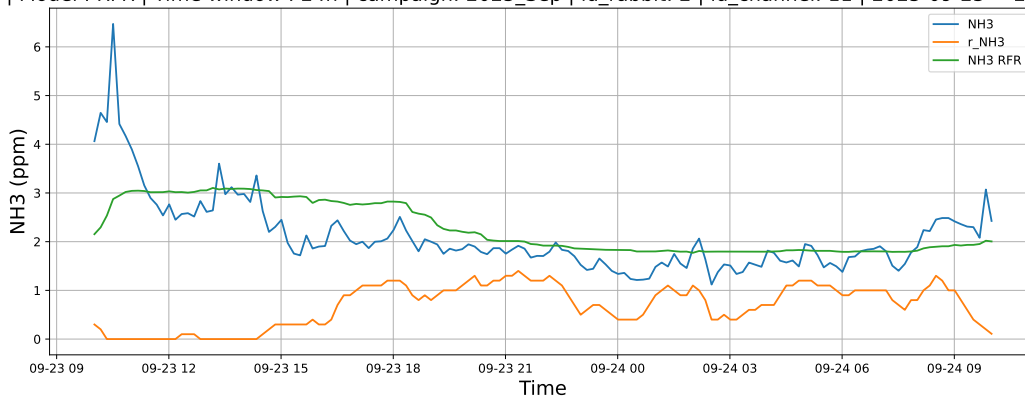
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

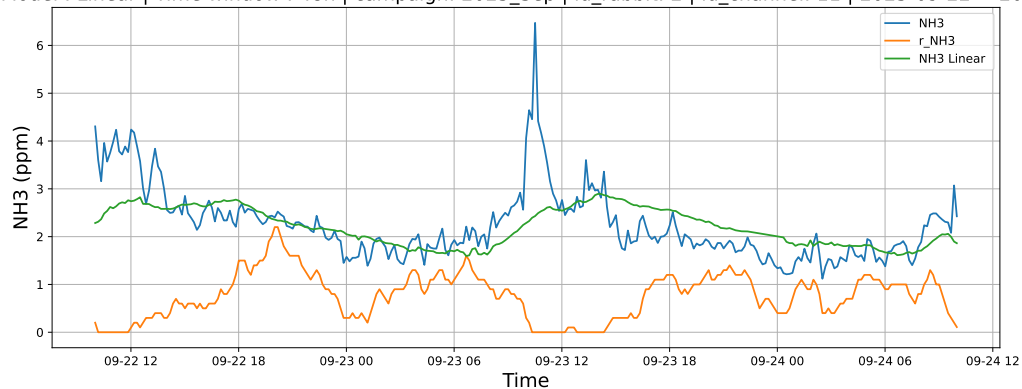


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

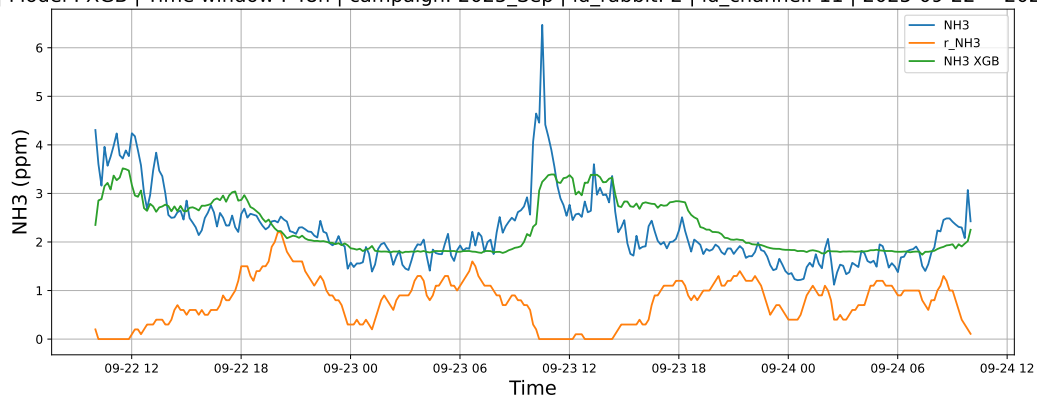


3.1.5.2 Time window : 48

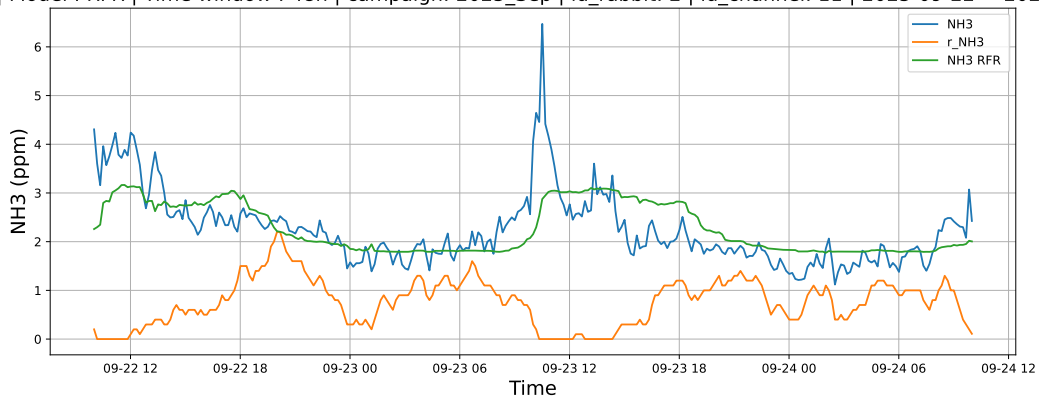
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

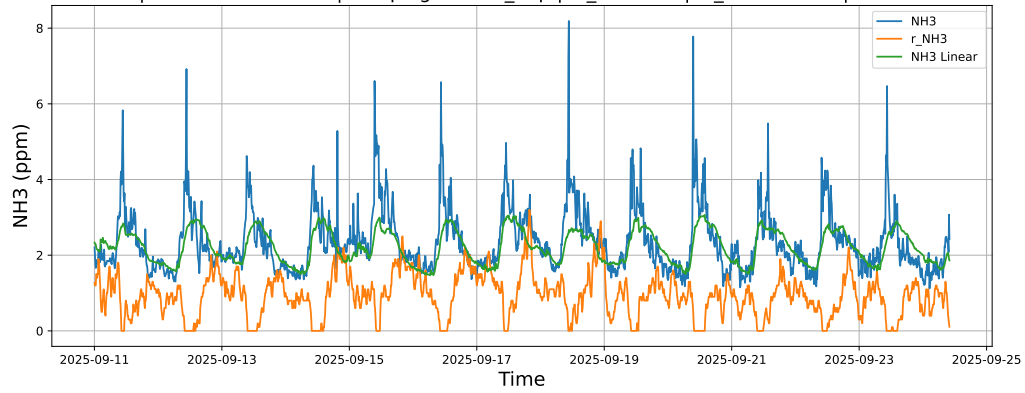


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

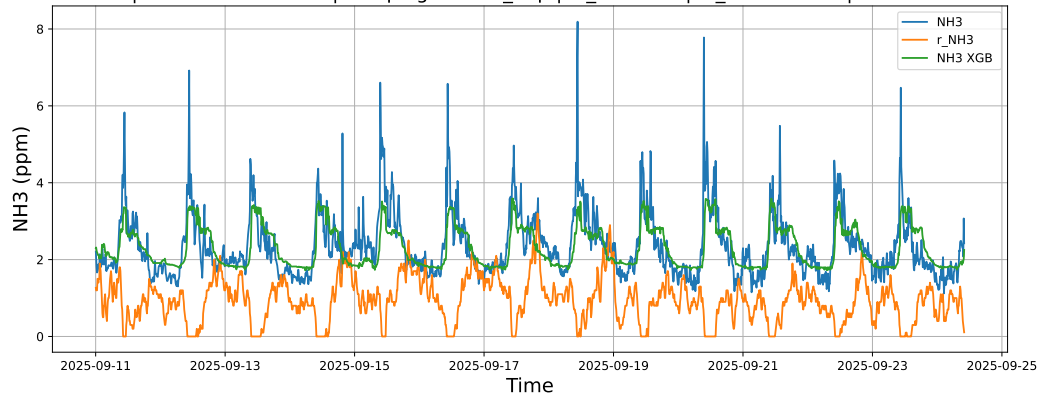


3.1.5.3 Time window : ALL

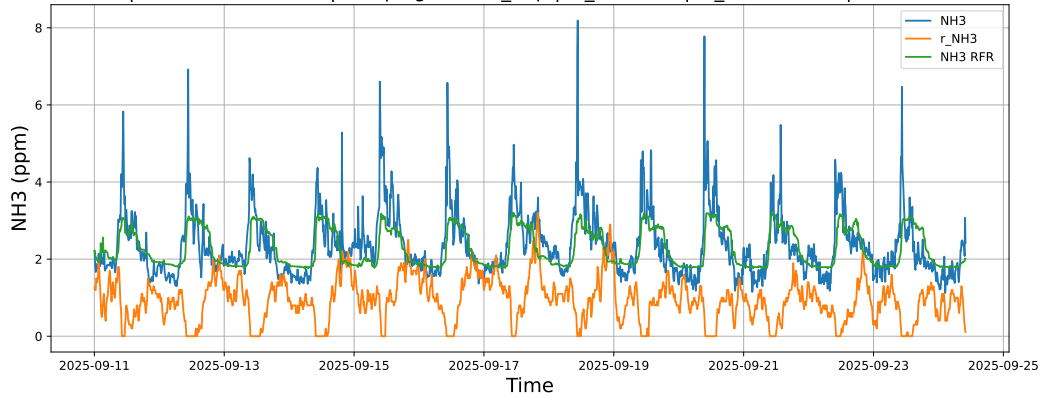
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



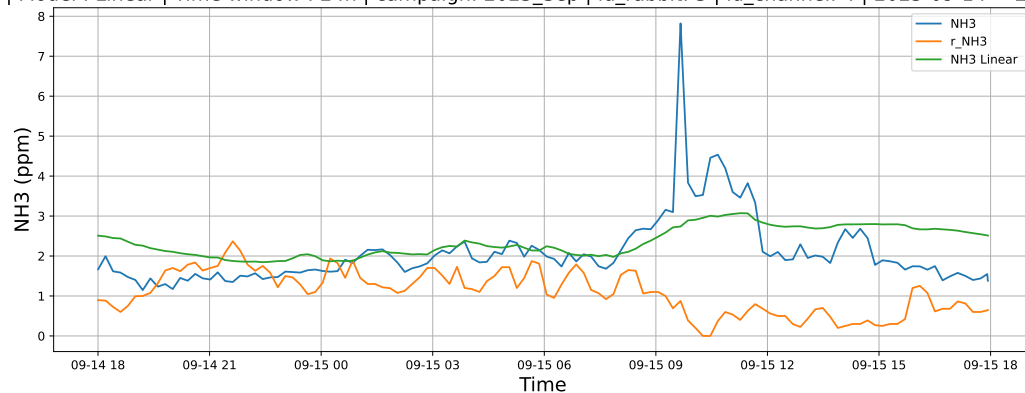
NH3 | Model : RFR | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



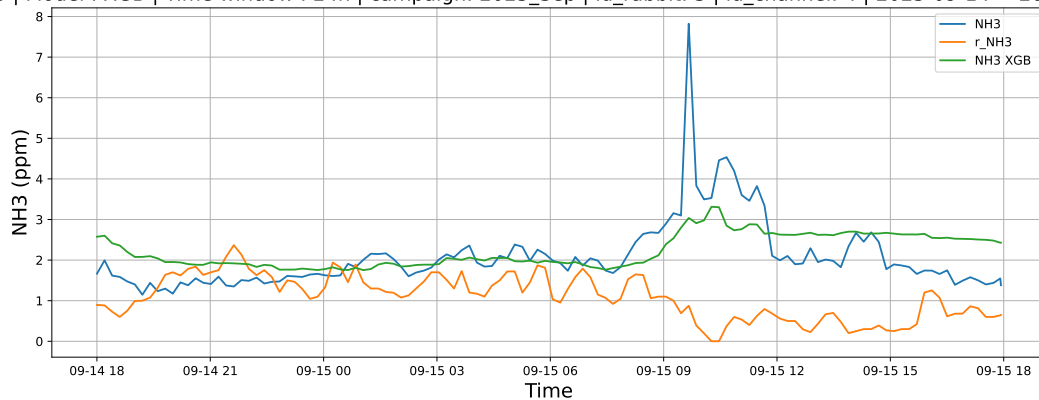
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

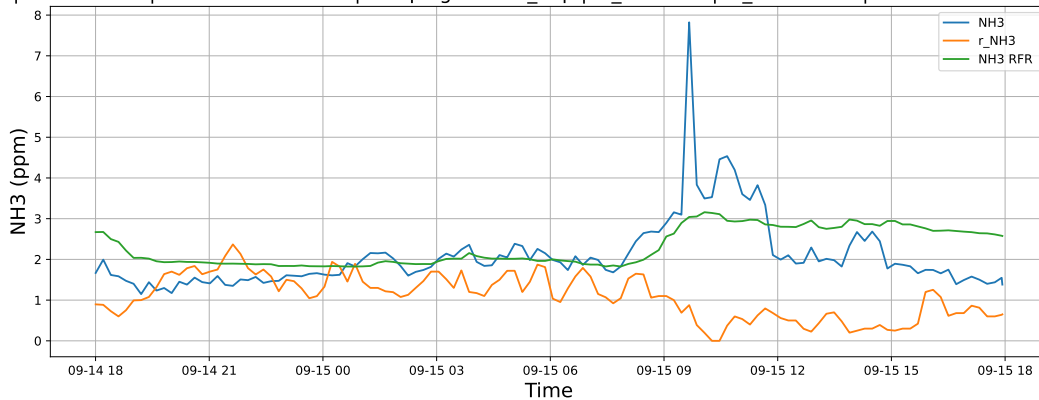
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

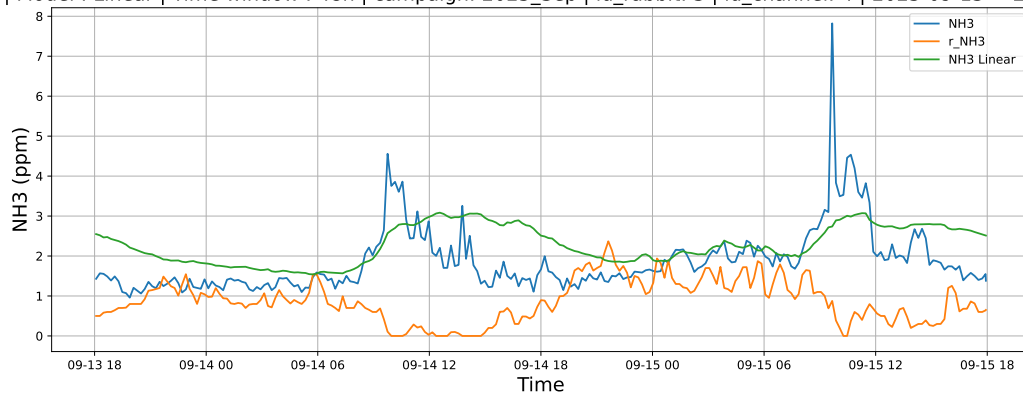


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

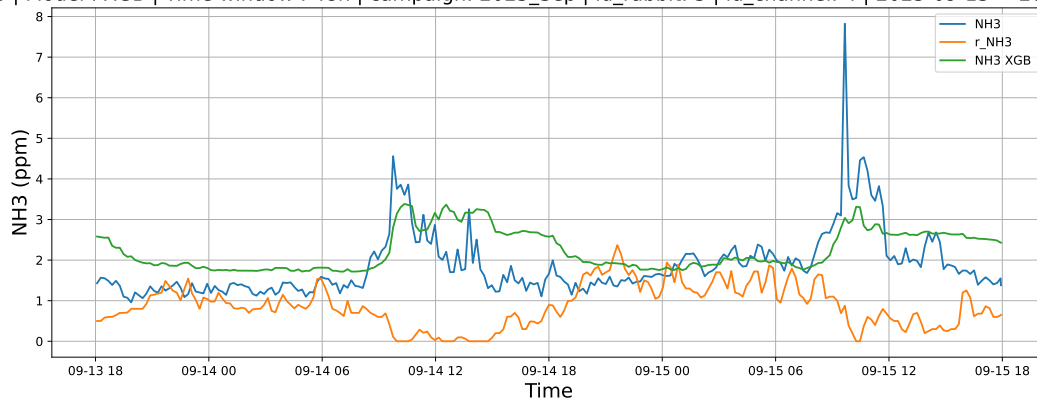


3.1.6.2 Time window : 48

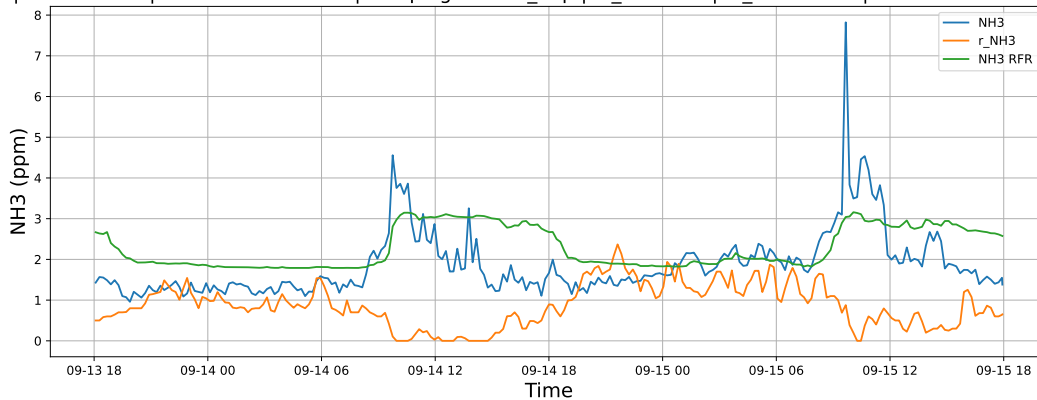
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

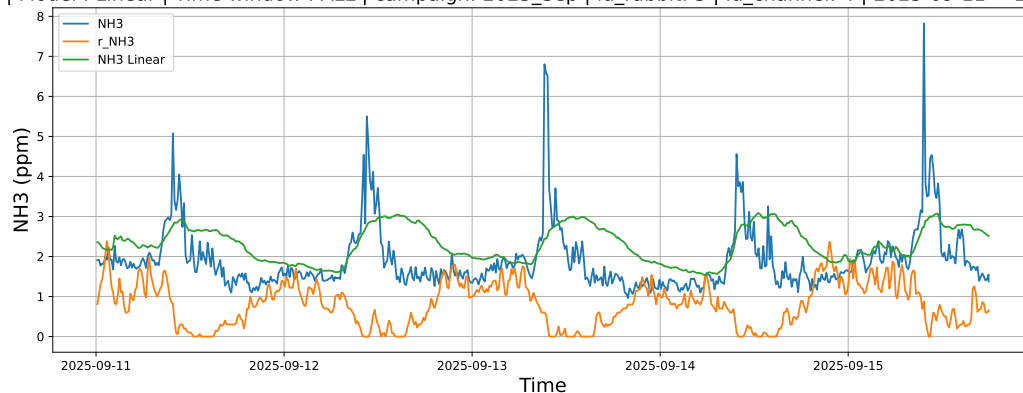


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

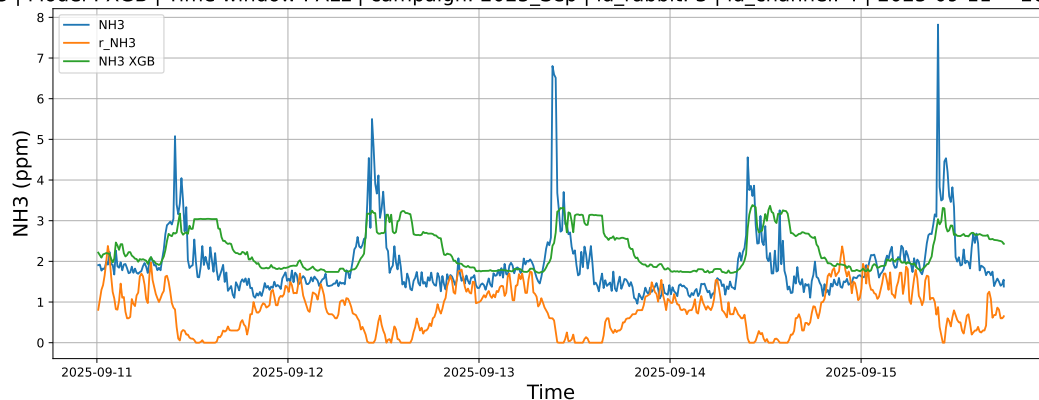


3.1.6.3 Time window : ALL

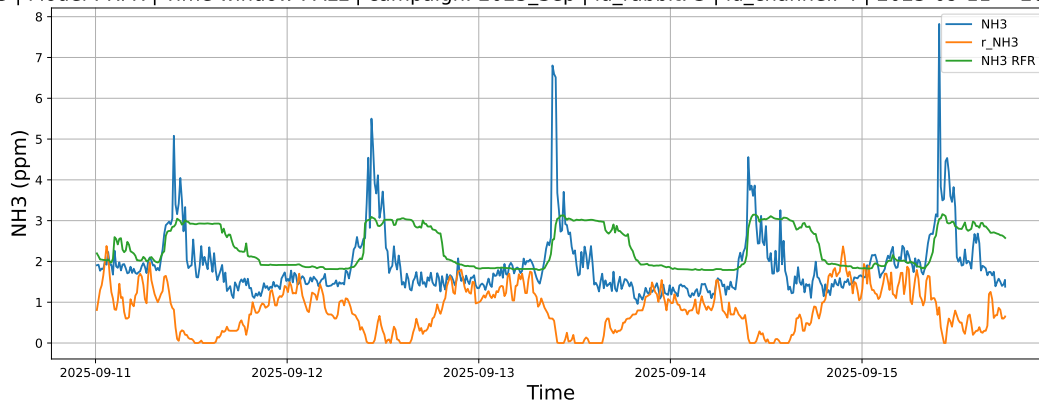
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



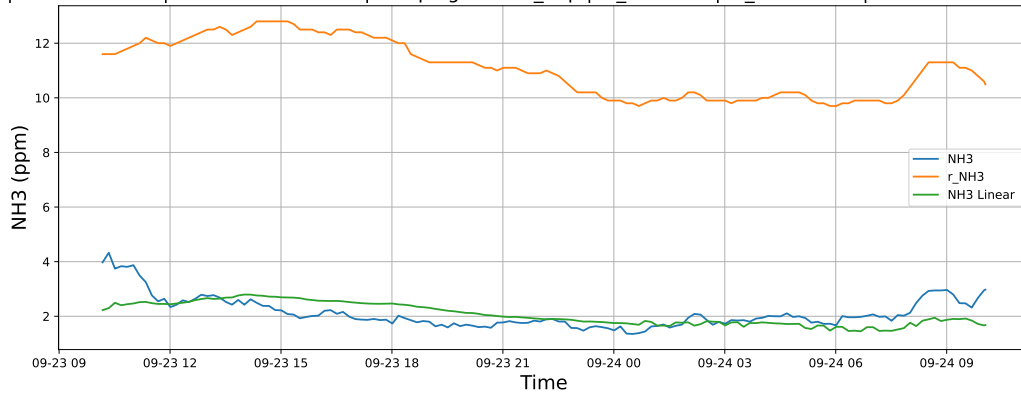
NH3 | Model : RFR | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



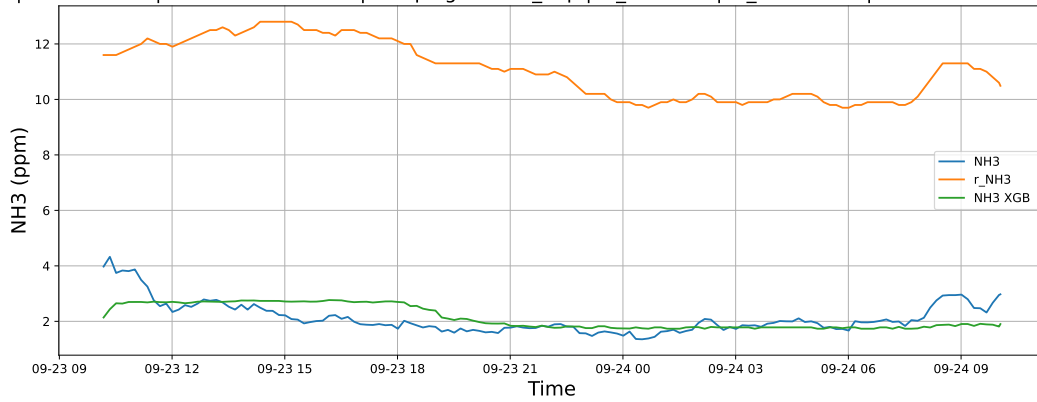
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

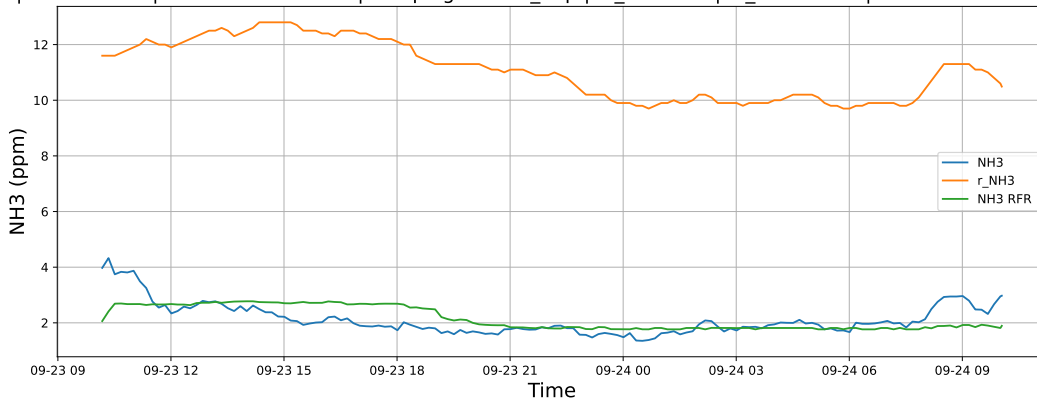
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

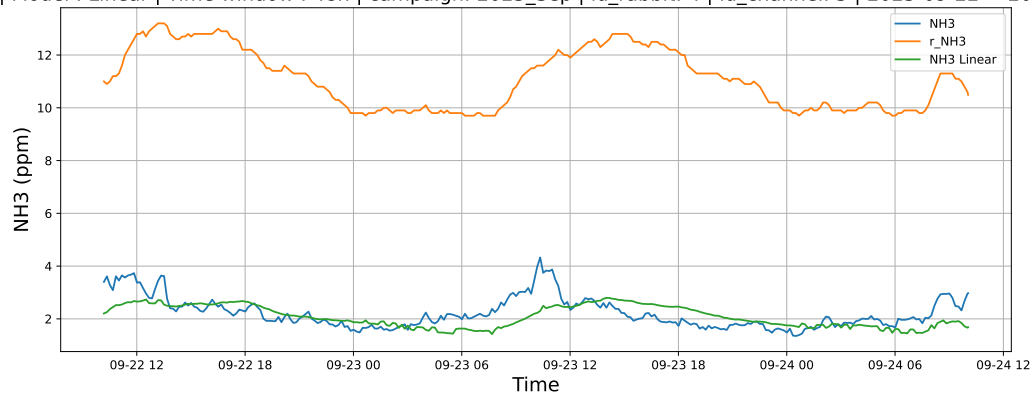


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

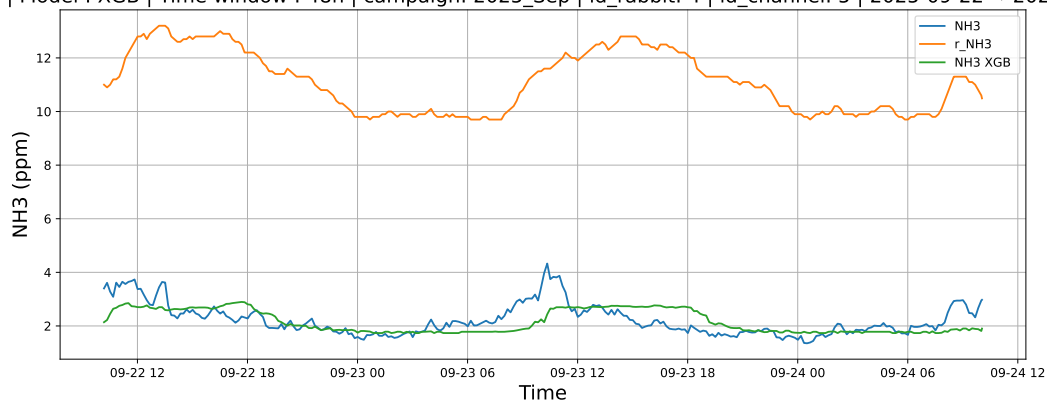


3.1.7.2 Time window : 48

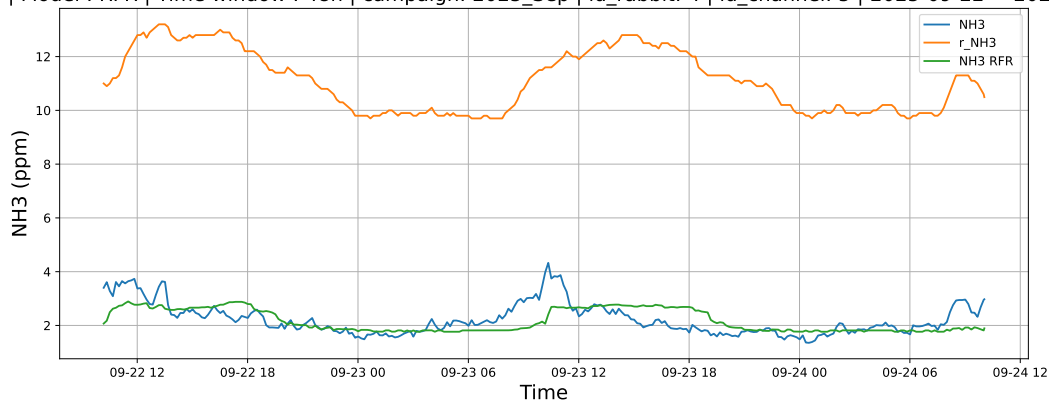
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

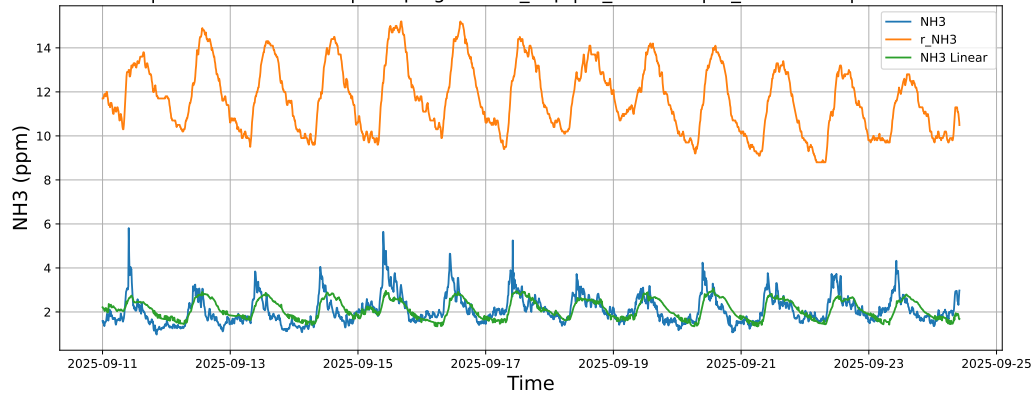


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

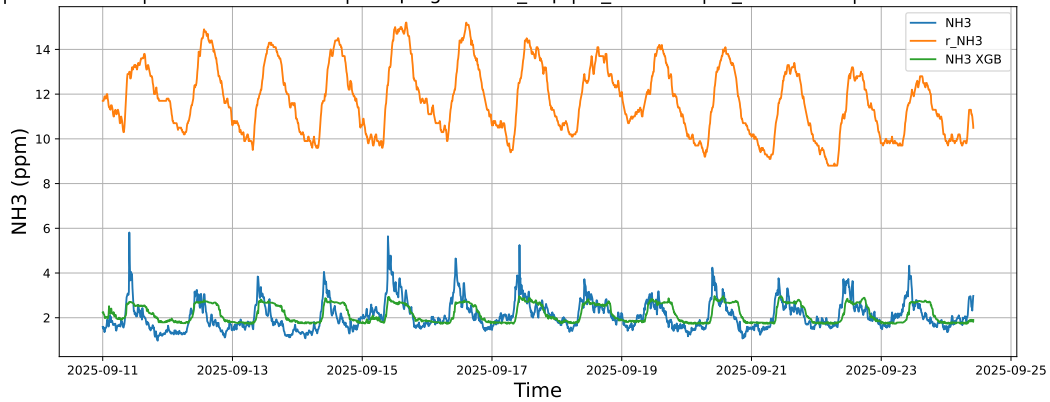


3.1.7.3 Time window : ALL

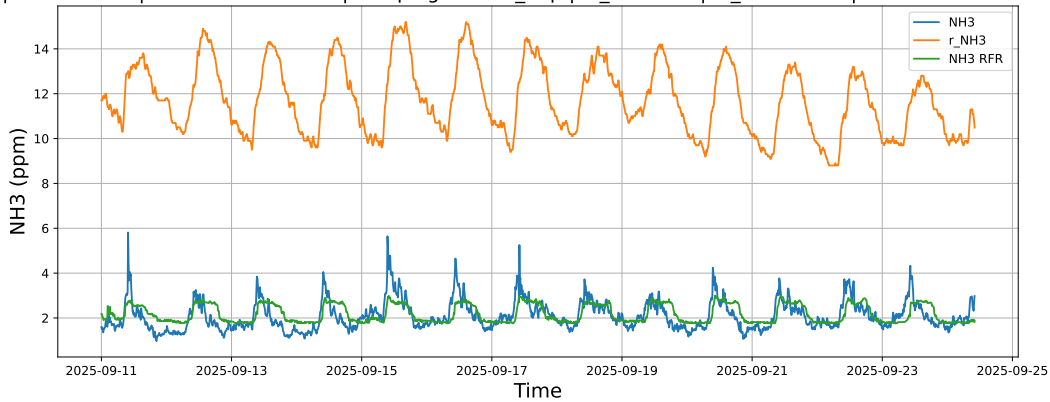
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



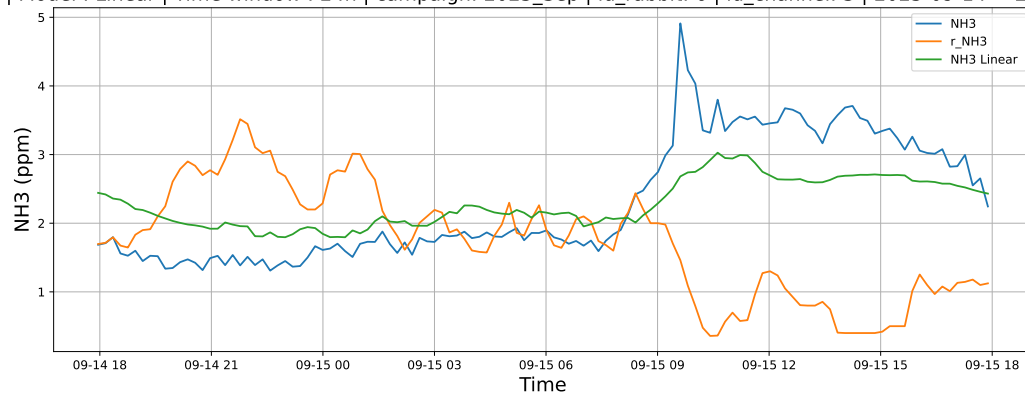
NH3 | Model : RFR | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



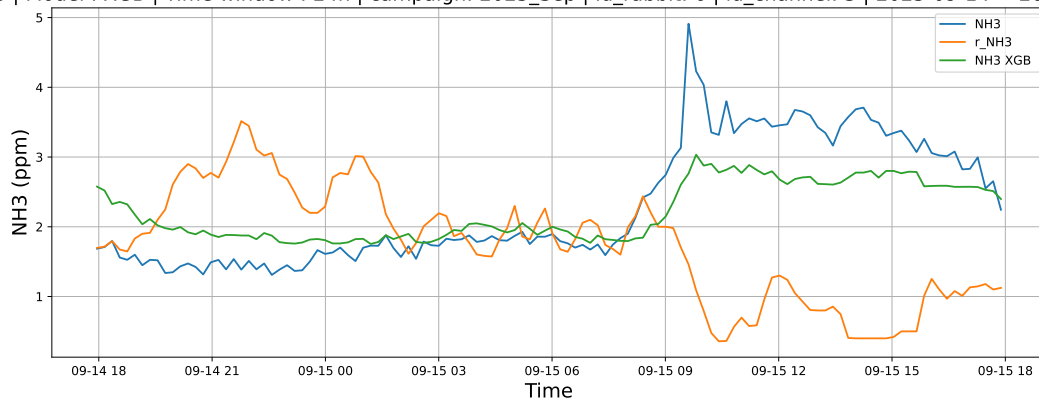
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

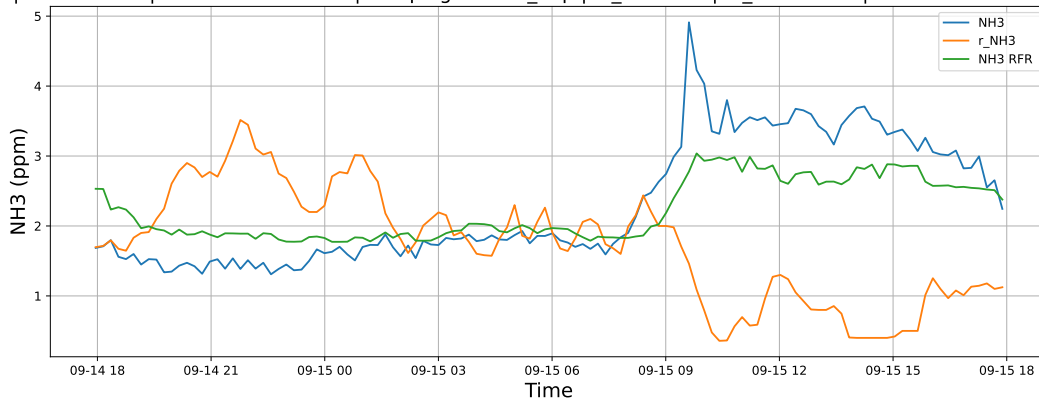
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : XGB | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

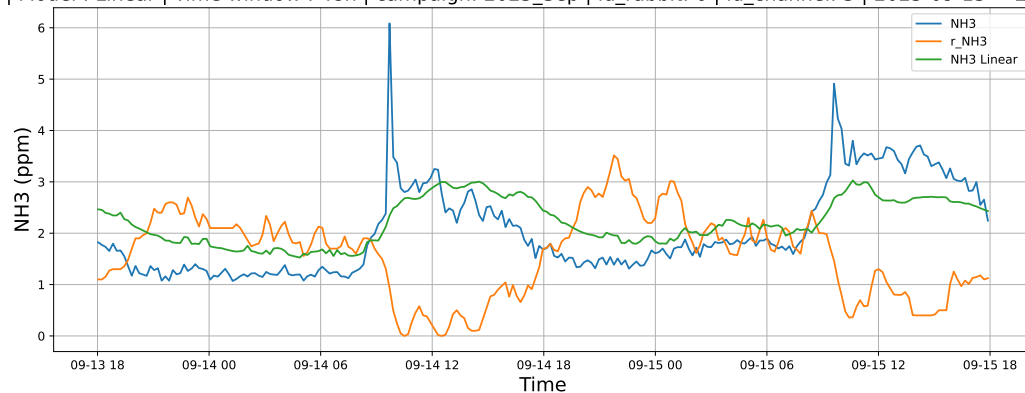


NH3 | Model : RFR | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

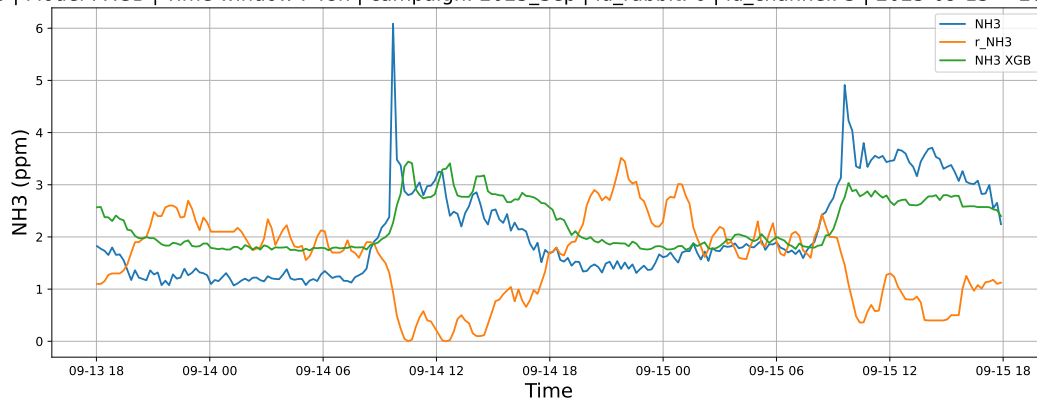


3.1.8.2 Time window : 48

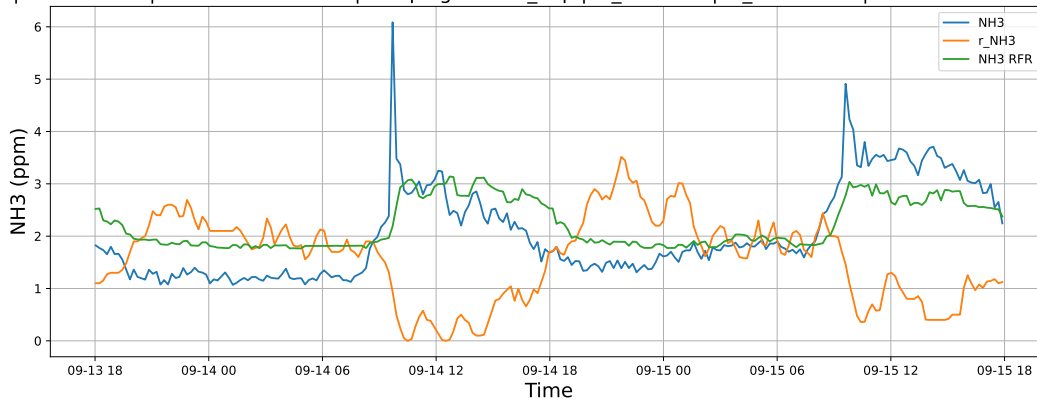
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : XGB | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

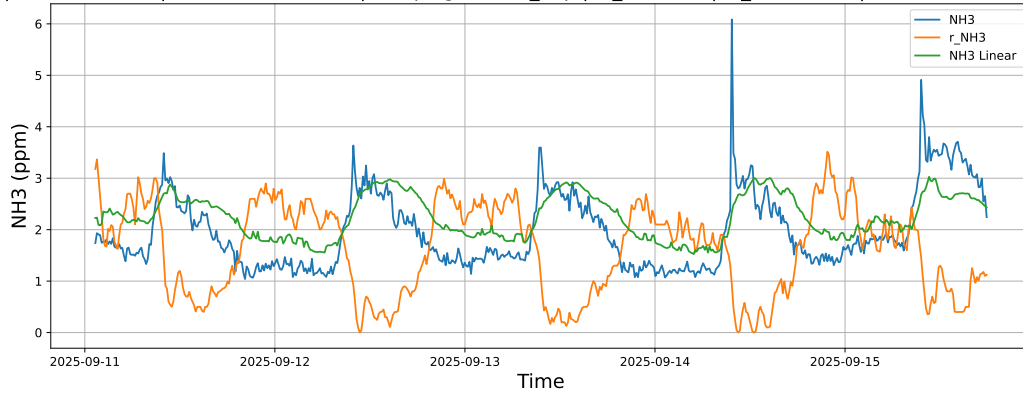


NH3 | Model : RFR | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

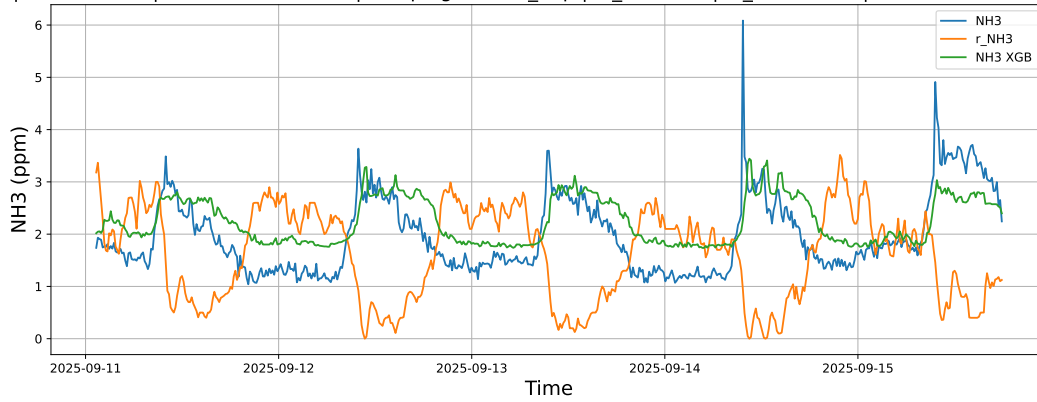


3.1.8.3 Time window : ALL

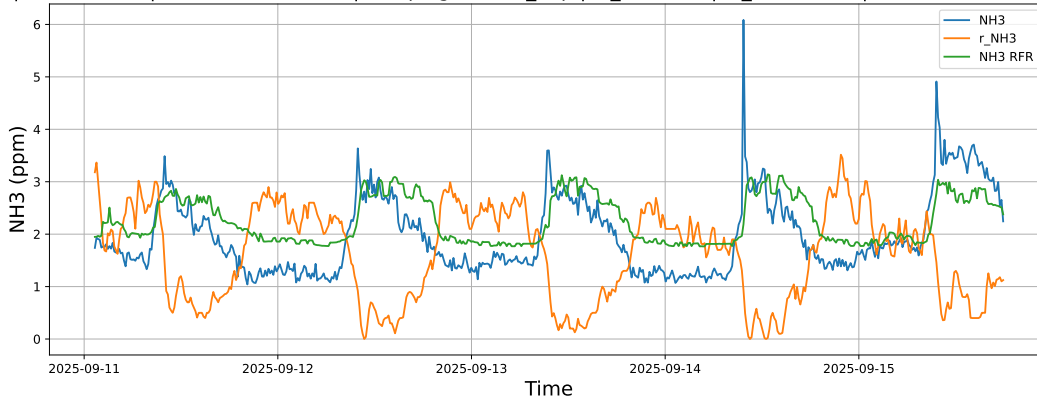
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : XGB | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : RFR | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

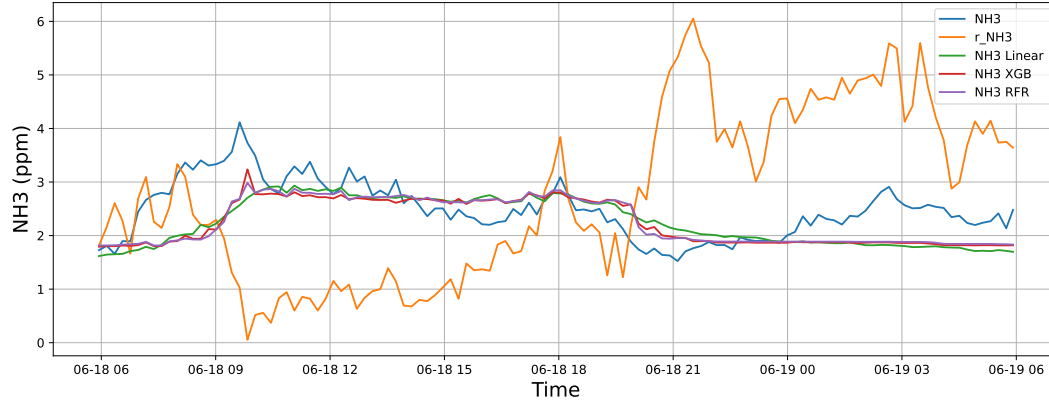


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

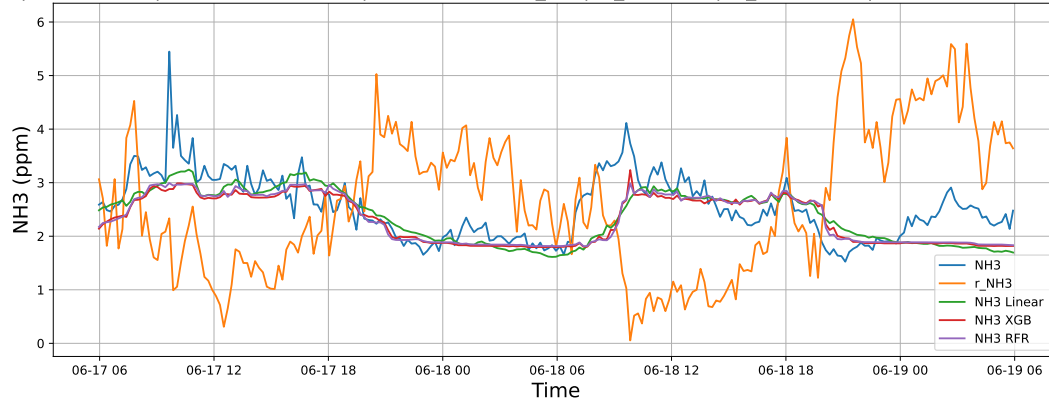
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



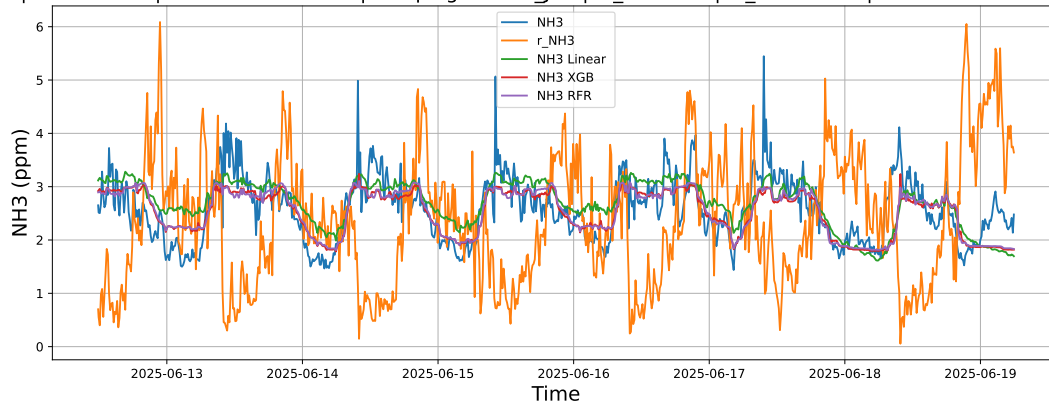
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

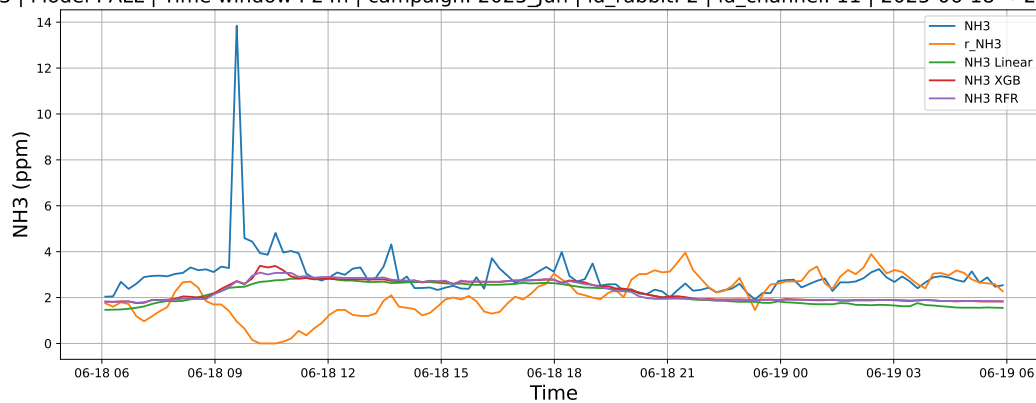
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

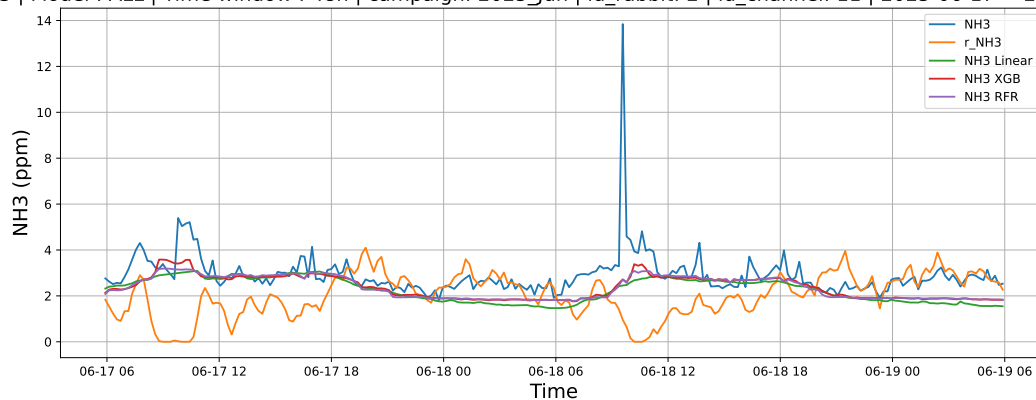
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



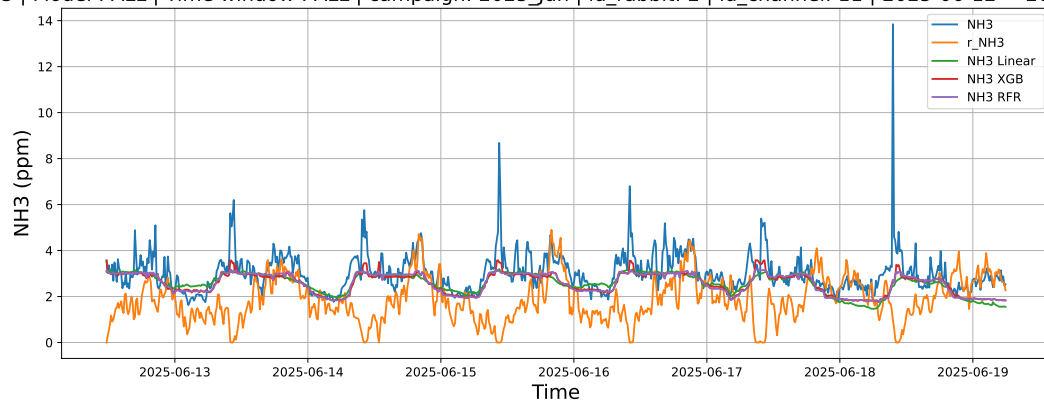
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

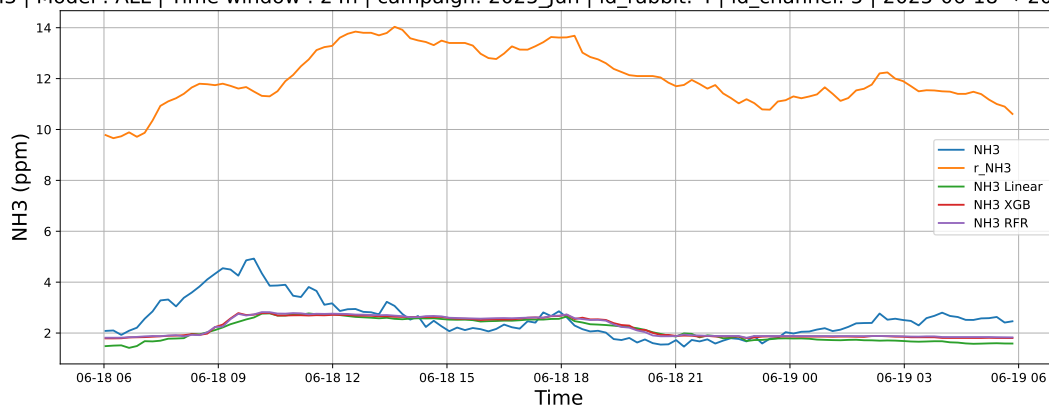
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

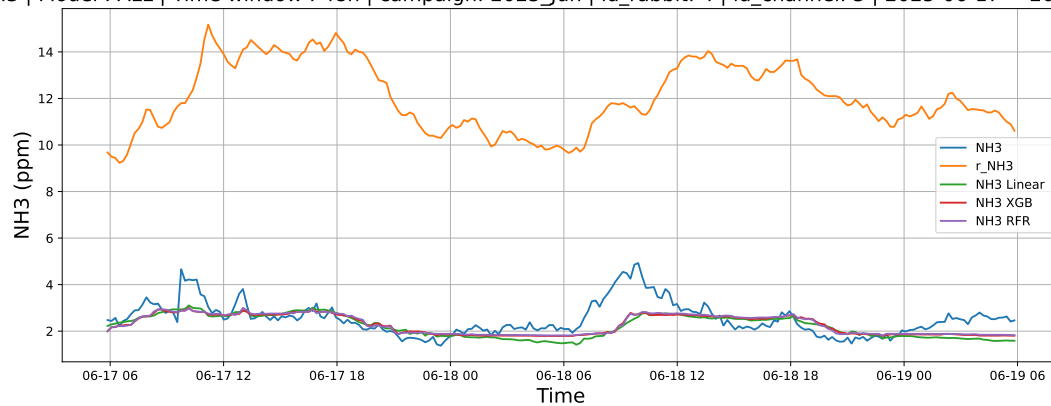
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



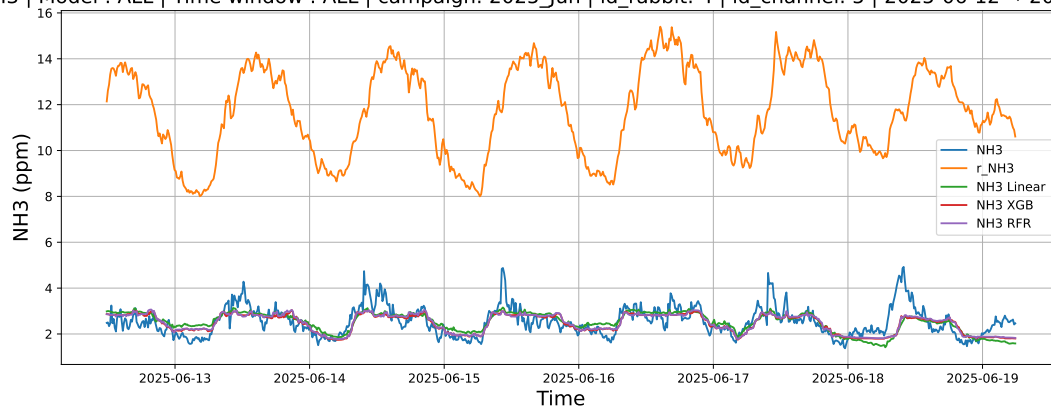
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

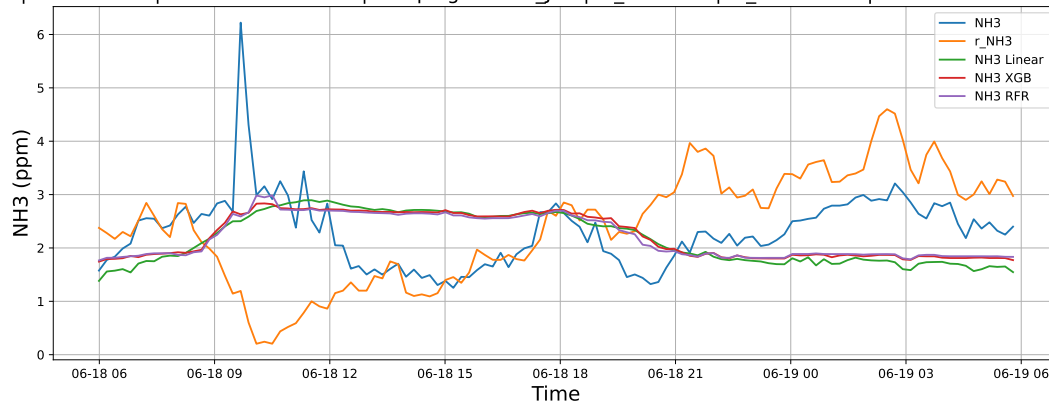
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

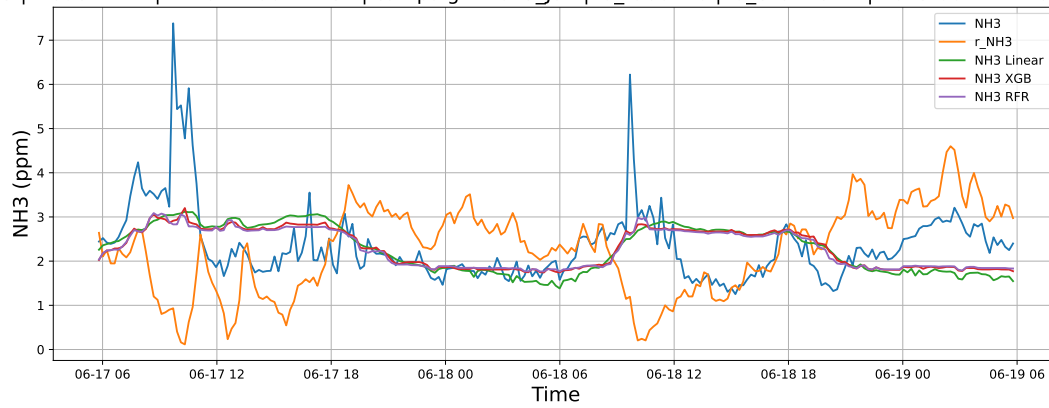
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



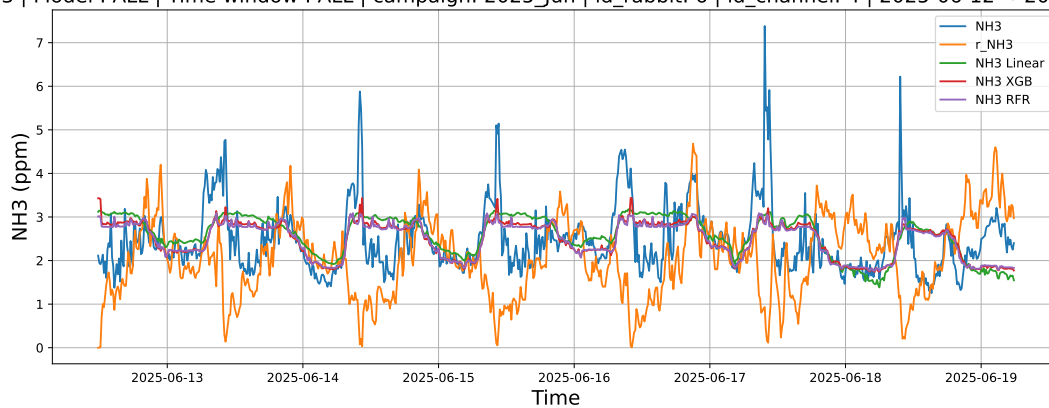
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

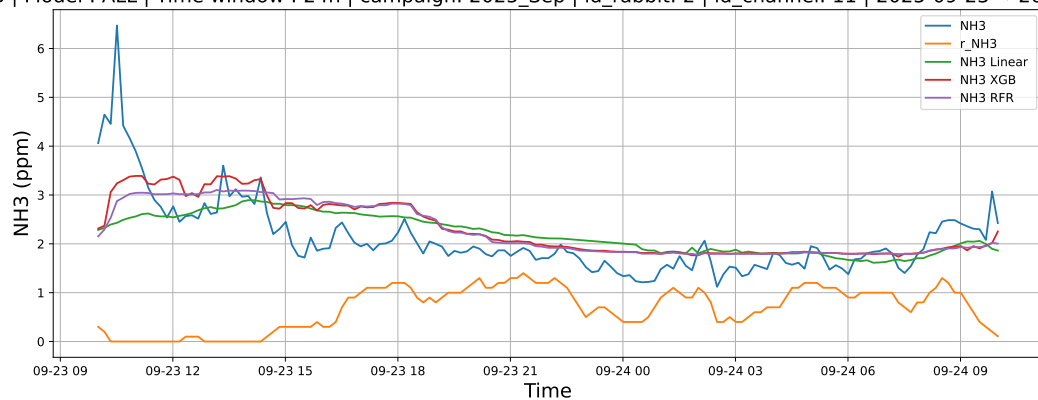
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

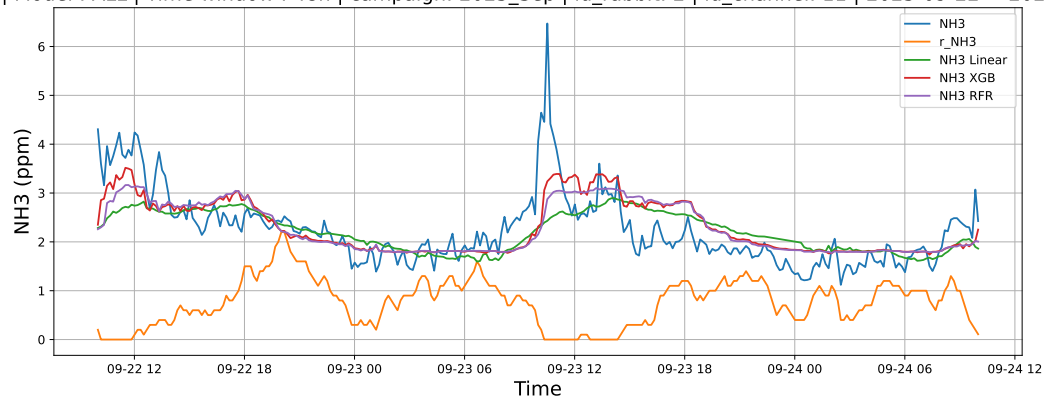
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



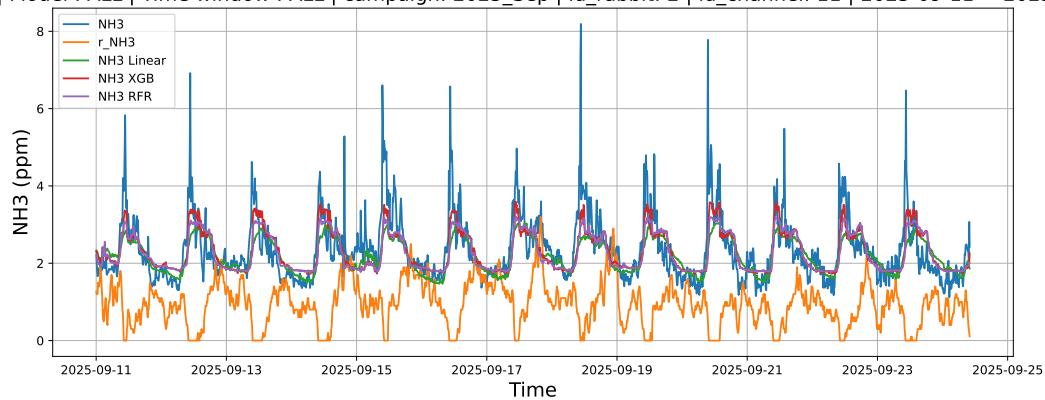
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

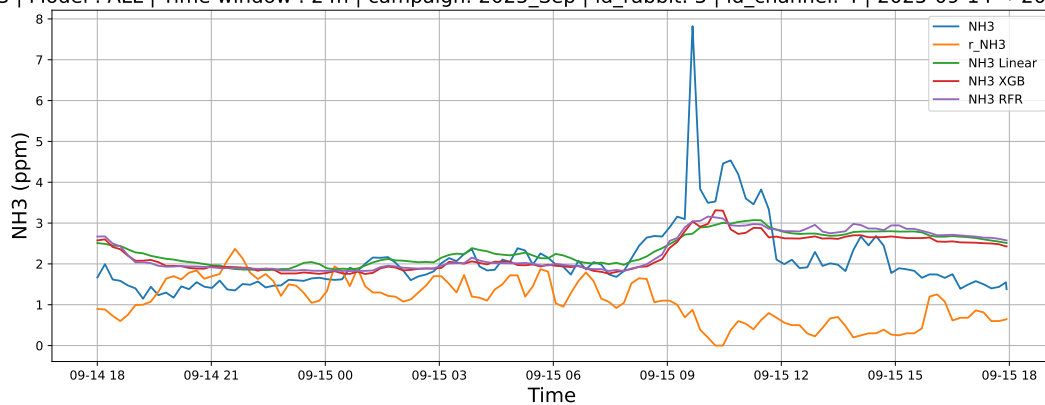
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

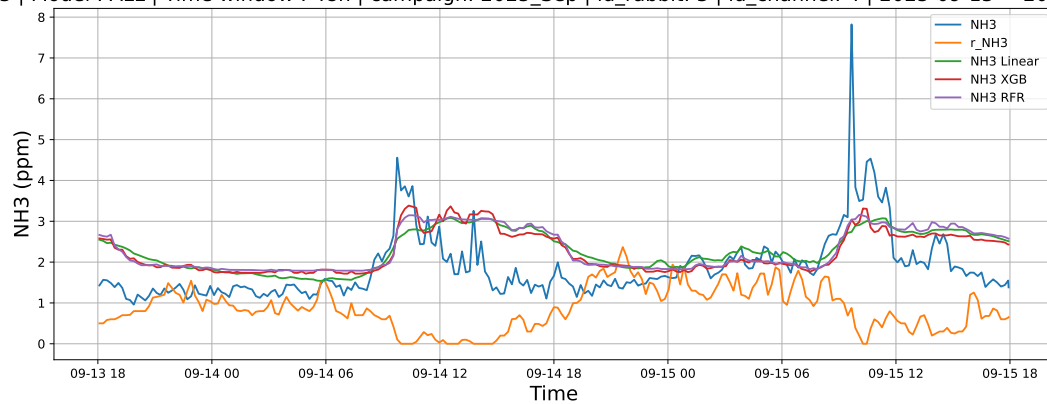
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



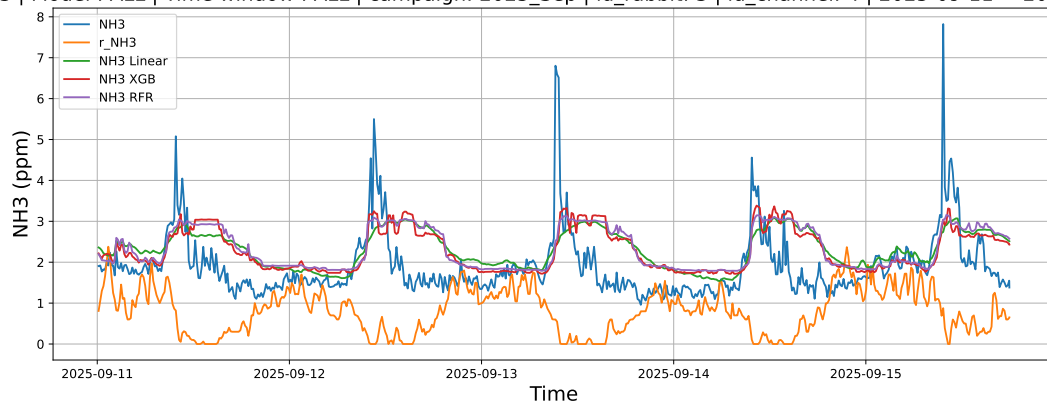
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

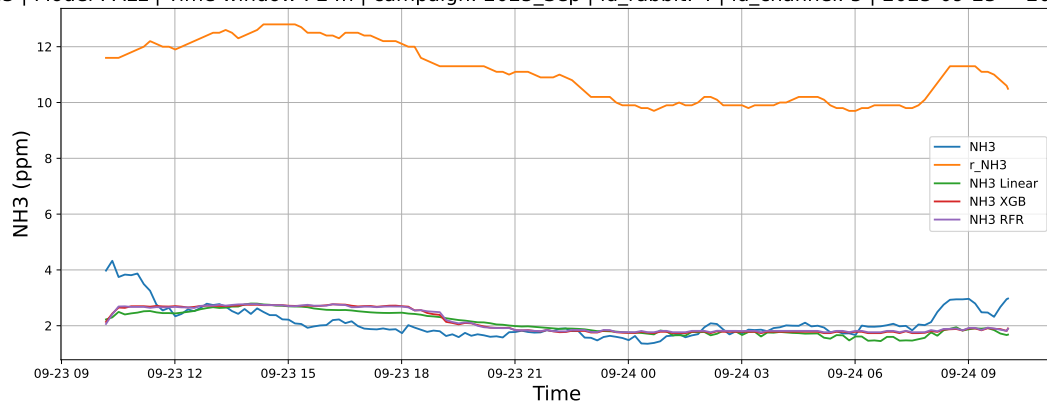
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

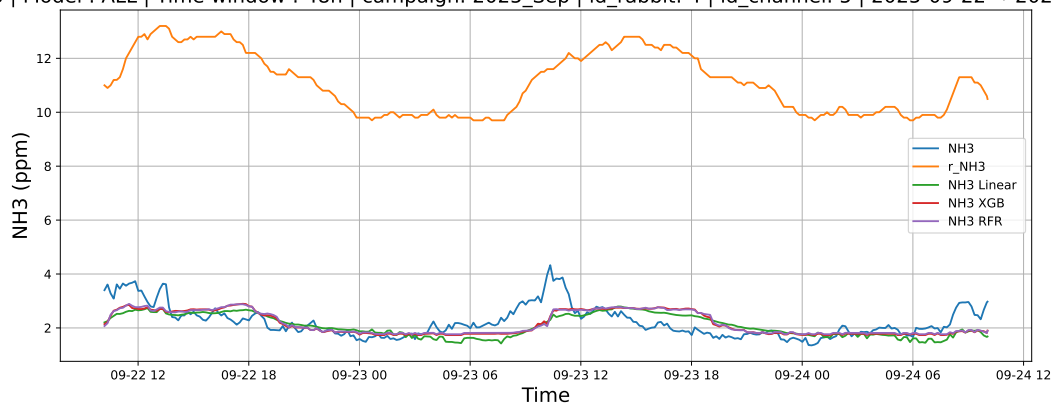
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



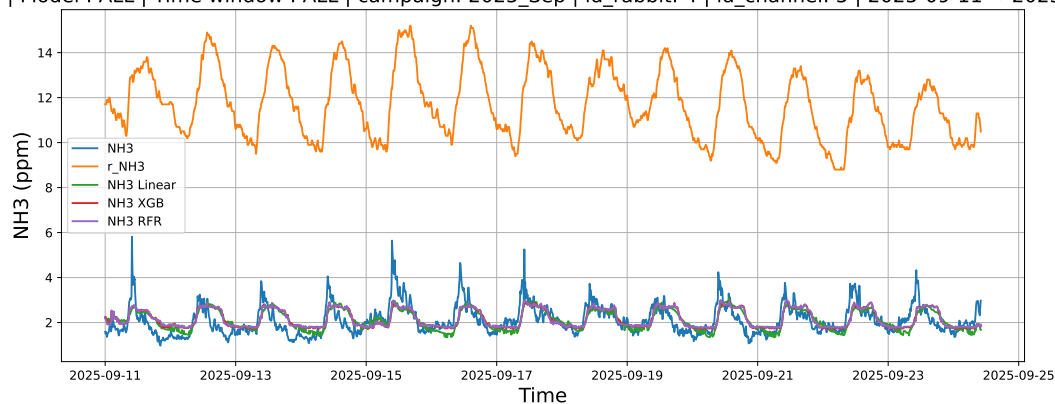
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

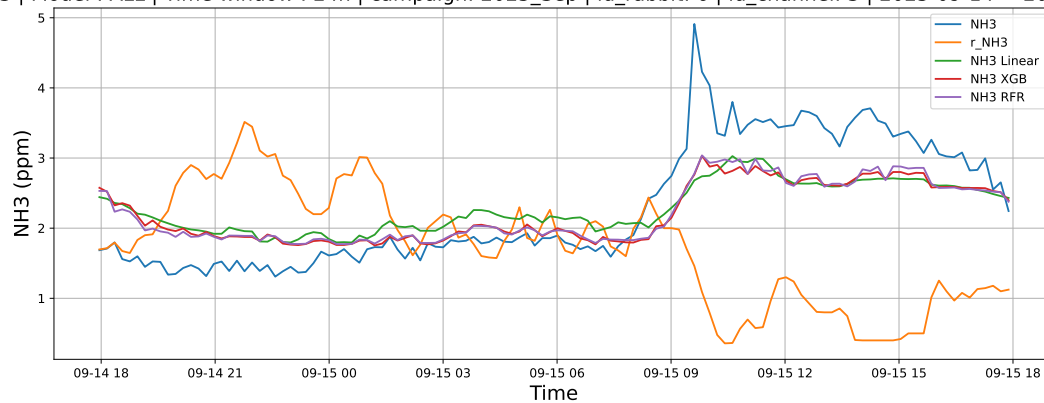
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

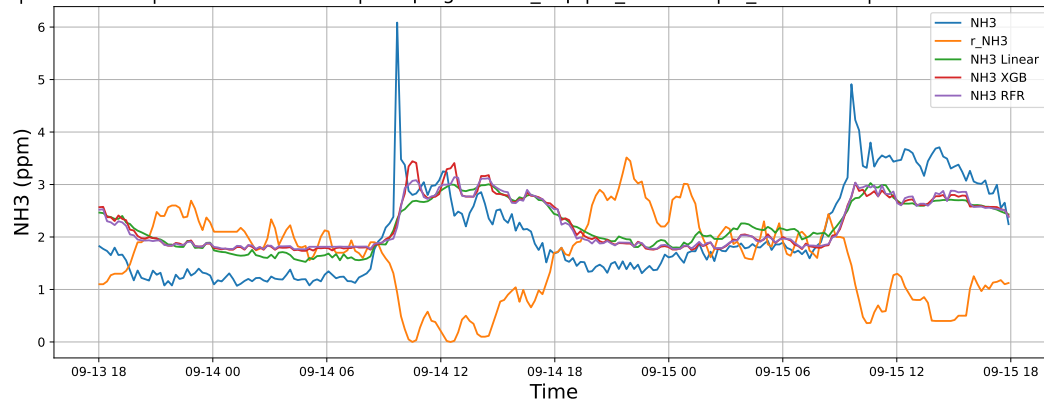
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

